Review of last week's lecture

Post-transcriptional events:

splicing;

Alternative splicing; self-splicing RNAs; tRNA splicing

capping: 0,1,2

Polyadenylation

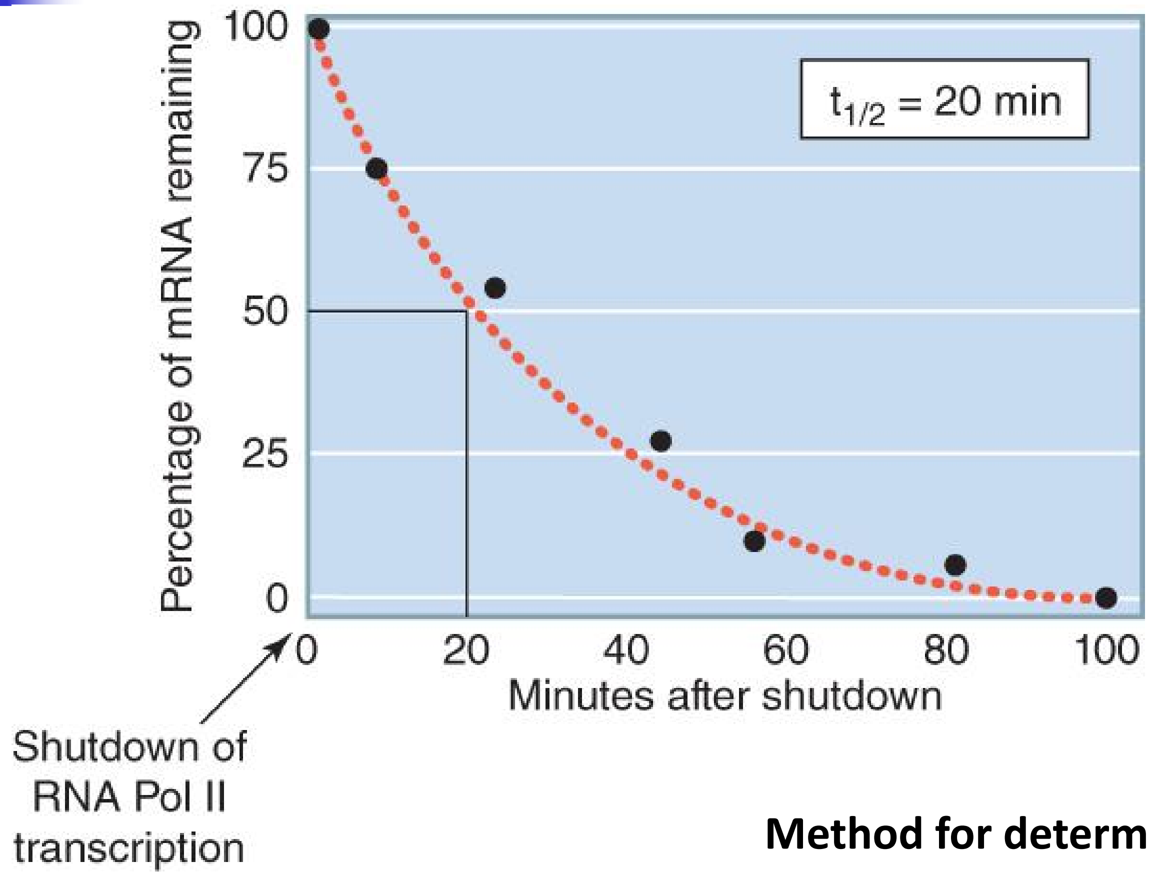
Other RNA Processing: rRNA and tRNA

Trans-Splicing; RNA editing

Posttranscriptional Control of Gene Expression

- mRNA stability
- Post-transcriptional gene silencing

Messenger RNAs are unstable molecules



Method for determining mRNA half-lives.

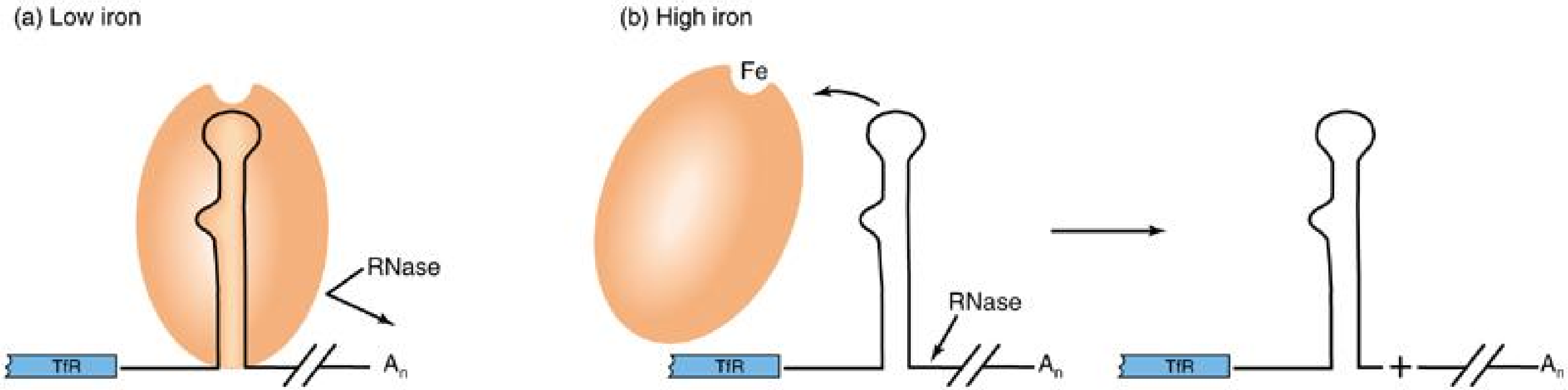
Casein mRNA stability

When mammary gland tissue is stimulated by prolactin, synthesis of casein protein increases dramatically

Most of this increase in casein is not due to increased rate of transcription of the casein gene - it is caused by an increase in half-life of casein mRNA

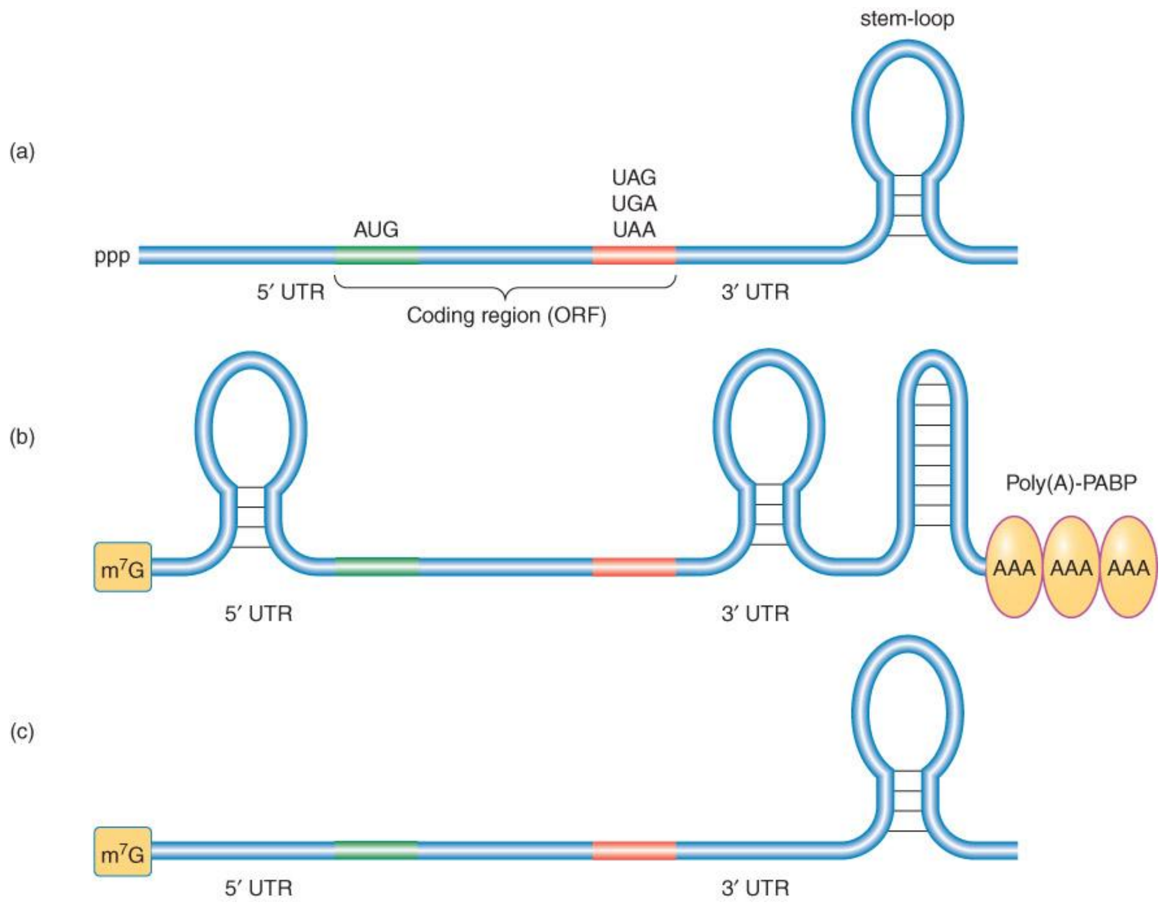
	RNA Half-life (h)	
Species of RNA	− Prolactin	+ Prolactin
rRNA	>790	>790
Poly(A)$^+$ RNA (short-lived)	3.3	12.8
Poly(A)$^+$ RNA (long-lived)	29	39
Casein mRNA	1.1	28.5

Destabilization of TfR mRNA by Iron



Aconitase apoprotein

Features of prokaryotic and eukaryotic mRNAs

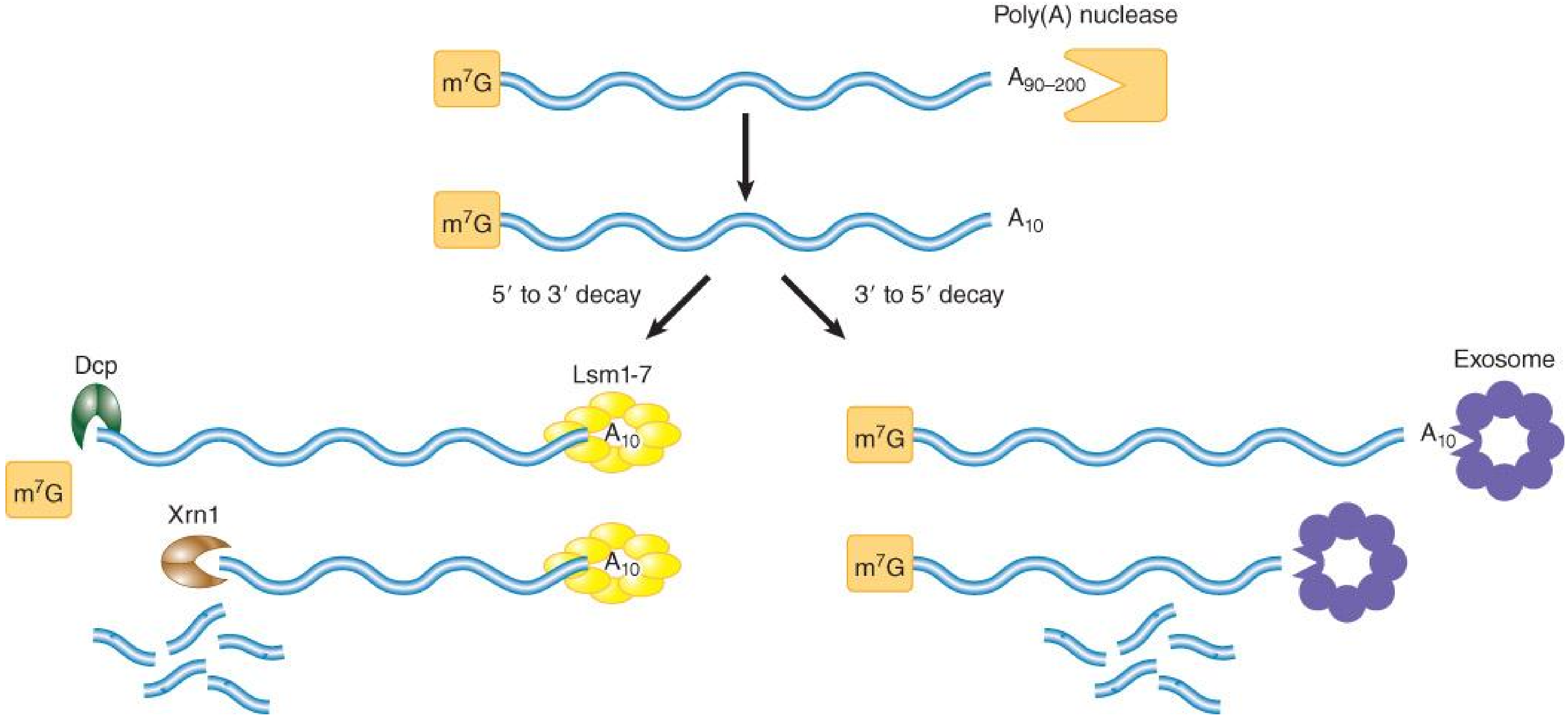


bacterial mRNA

eukaryotic mRNA

Some histone mRNA

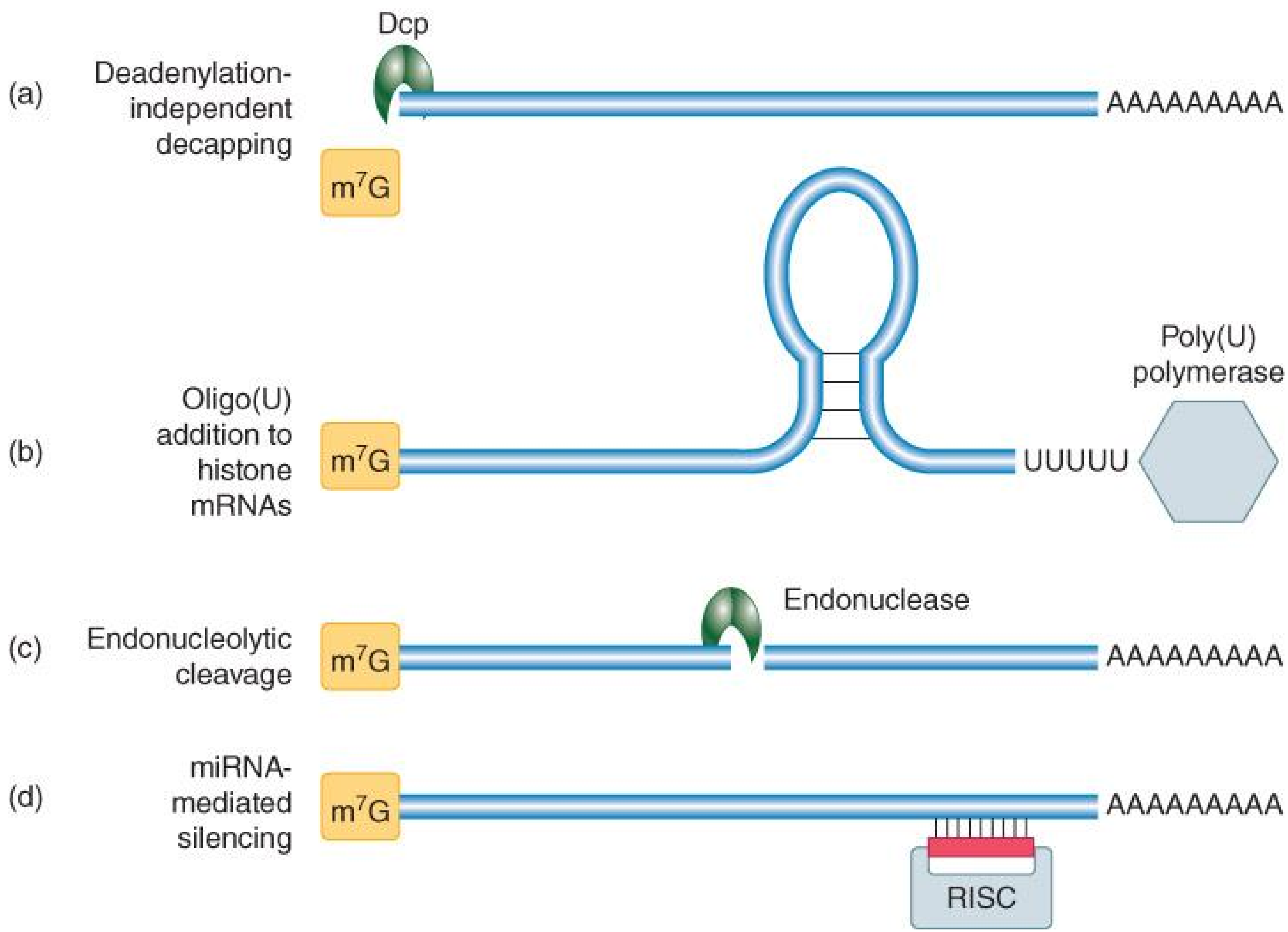
The two major mRNA decay pathways are deadenylation-dependent in eukaryotes



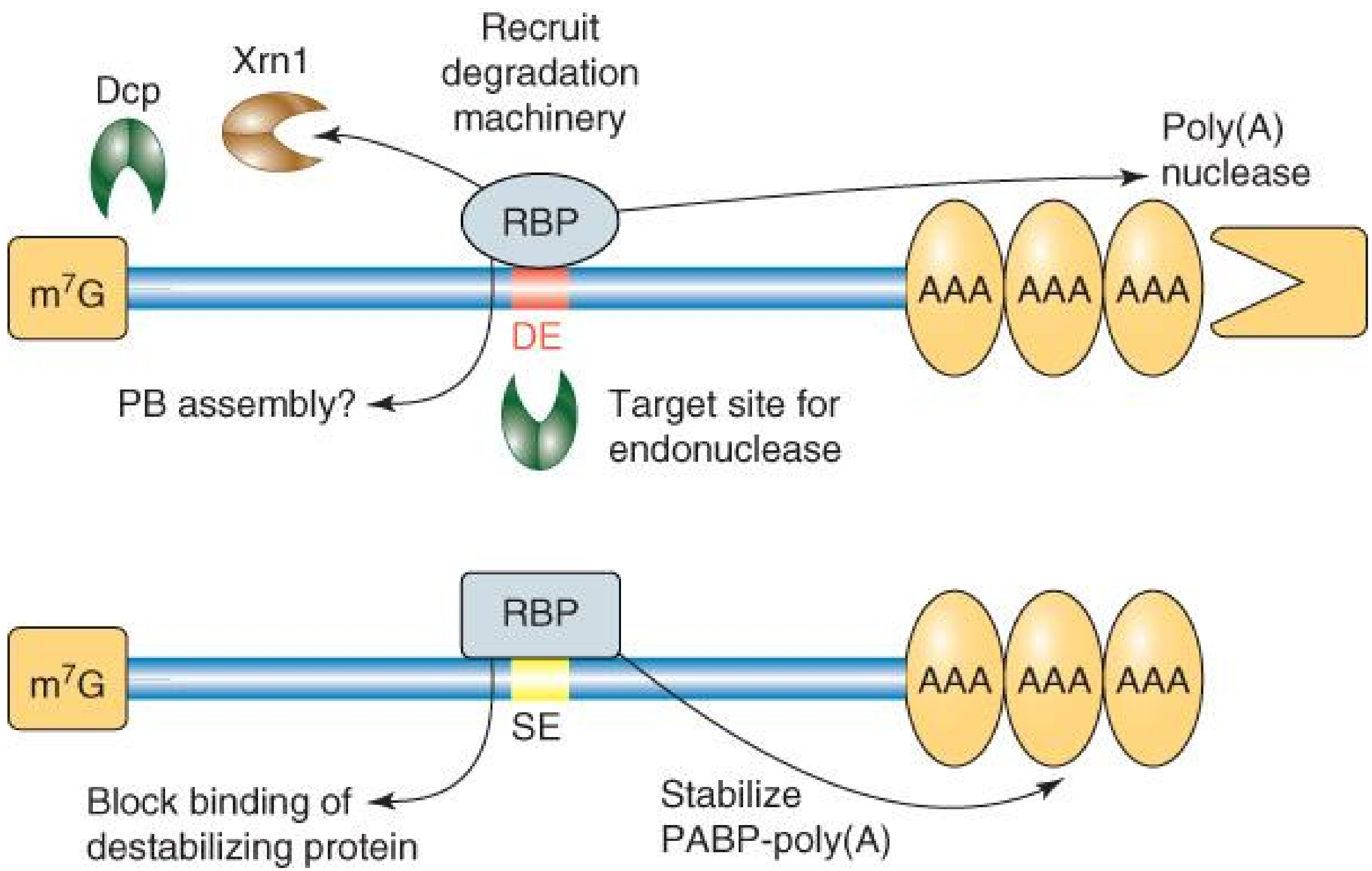
Decapping enzyme Dcp;
Xrn1, 5' to 3' exonuclease

Endoribonuclease – A ribonuclease that cleaves an RNA at an internal site(s).
Exoribonuclease – A ribonuclease that removes terminal ribonucleotides from RNA.

Other decay pathways in eukaryotic cells



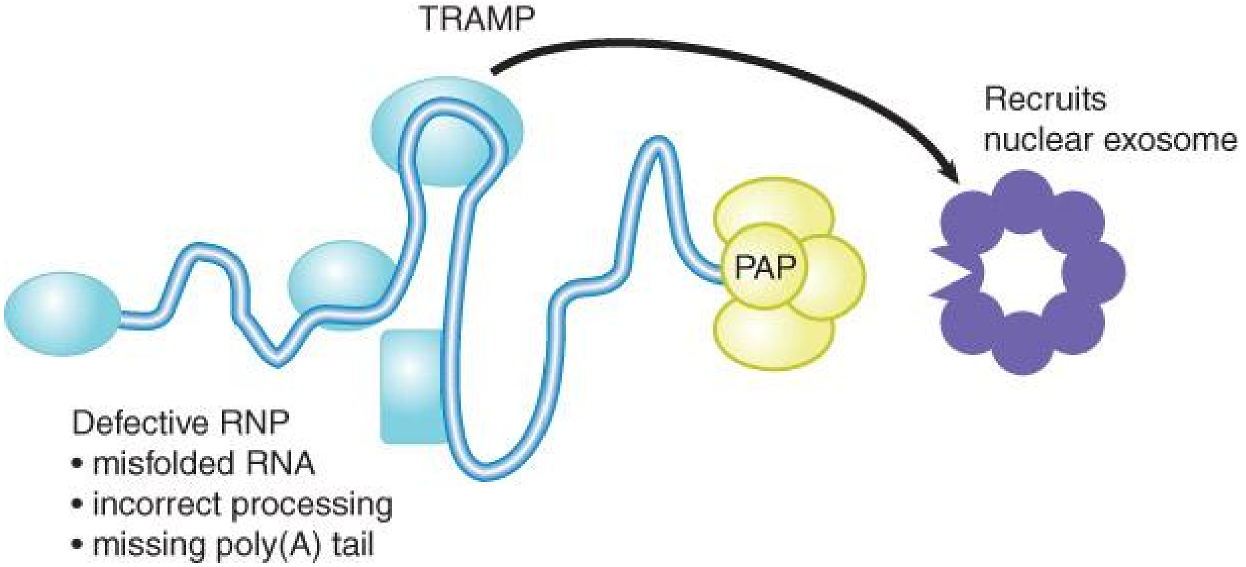
mRNA-specific half-lives are controlled by sequences or structures within the mRNA



Destabilizing element (DE)
Stabilizing element (SE)

Newly synthesized RNAs are checked for defects via a nuclear surveillance system

- Aberrant nuclear RNAs are identified and destroyed by an **RNA surveillance system**.
- The nuclear exosome functions both in the processing of normal substrate RNAs and in the destruction of aberrant RNAs.
- The yeast **TRAMP** (Trf4/Air2/Mtr4 polyadenylation) complex recruits the exosome to aberrant RNAs and facilitates its 3′ to 5′ exonuclease activity.



Post-transcriptional Gene silencing (RNAi)

Types of RNA

- Coding RNA: messenger RNA (mRNA)
- Non-coding RNA:
  - **Ribosomal RNA (rRNA)**
  - **Transfer RNA (tRNA)**
  - **Small nuclear RNA (snRNA)**
  - **Small nucleolar RNA (snoRNA)**

  - **Small interfering RNA (siRNA)**
  - **Micro RNA (miRNA)**
  - **Others (long non-coding RNA；NATs etc.)**

Relatively new types of RNA

- siRNA: active molecules in RNA interference; degrades mRNA (act where they originate)
- Piwi-interacting RNA (piRNA): piRNA forms RNA-protein complexes through interactions with piwi proteins. Gene silencing of retrotransposons and other genetic elements in animal cells
- miRNAs: tiny 21–24-nucleotide RNAs; acting as transcript cleavage or translational regulators of protein-coding mRNAs (regulate elsewhere)
- Long non-coding RNA (lncRNA) : non-protein coding transcripts longer than 200 nucleotides
- circRNA: circular RNA

Glossary

- **Sense RNA:** a (shorter) version of a particular mRNA strand (single stranded).
- **antisense RNA:** a single stranded RNA complementary to a particular mRNA sequence; **NAT**: Natural antisense transcript
- **dsRNA:** double stranded RNA
- **RNA-dependent RNA-polymerase (RdRp):** transferase class enzyme that catalyzes the template-directed step by step addition of ribonucleotides, using a single-stranded RNA template
- **RISC:** RNA induced silencing complex
- **VIGS:** virus-induced gene silencing

So, what is RNAi ?

- **PTGS** : post-transcriptional gene silencing, is a transcriptional surveillance mechanism leading to sequence-specific degradation of mRNA. It is caused by dsRNA

- **Why PTGS?:** A self defense mechanism evolved to target infection by viruses and transposons potential damage

So, what is RNAi ?

• An evolutionally highly conserved process of post-transcriptional gene silencing (**PTGS**) by which double stranded RNA causes sequence-specific degradation of mRNA sequences

→ **a transcriptional surveillance mechanism**

In brief

- **RNA interference (RNAi):** The phenomenon of gene silencing caused by dsRNA

What for ?

• A self defense mechanism evolved to target infection by viruses and transposons potential damage

The History of RNAi

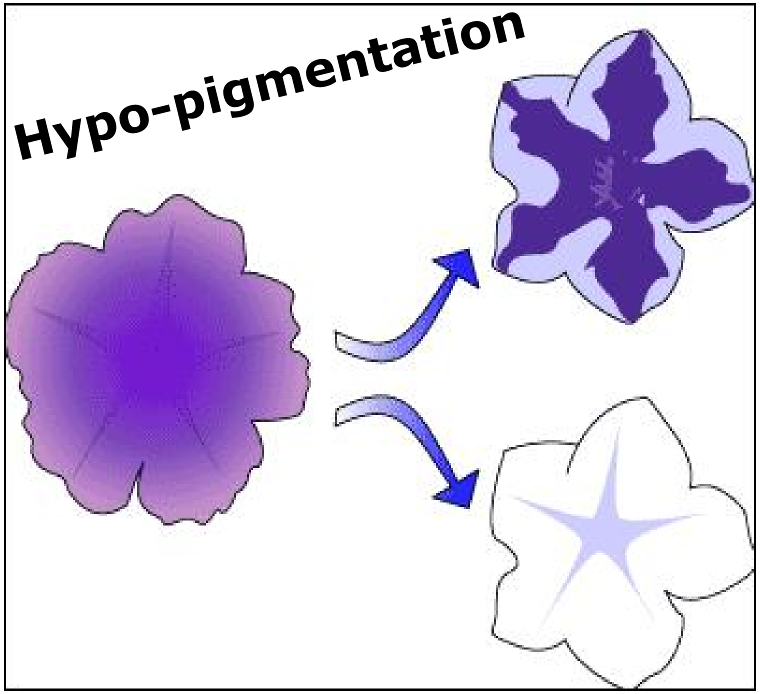
Gene-specific inhibition of expression by anti-sense nucleic acids was discovered 30 years ago (Inouye, 1988)

Sense RNA have Similar Effect

- 1990, in petunias (Rich Jorgensen)

- Tried to deepen the color by boosting the activity of the pigment producing gene

The result was…



- **<u>co-suppression</u>** silencing of an endogenous gene by the presence of a homologous transgene. (expression of both genes is suppressed)

- **<u>Quelling</u>** – in fungi, (Neorospora carassa) by the introduction of a transgene

The Discovery of RNAi in Animal

As always, it was an accident…

1995 (Gou & Kemphues) in C. elegans – *PAR1* gene function

Sense and Antisense are sufficient to cause interference

- Silencing effects are evident in both injected animals and their progeny even though RNA transcripts are rapidly degraded in the early embryo

For the next 3 years -

“The basis for the sense effect is under investigation…”

Until **1998** in C.elegans (Fire & Mello) –

native RNA vs. interfering RNA

The **dsRNA** that they inadvertently created inhibited homologous mRNA expression **much more strongly** than either the sense or the anti-sense strands alone (!!??).

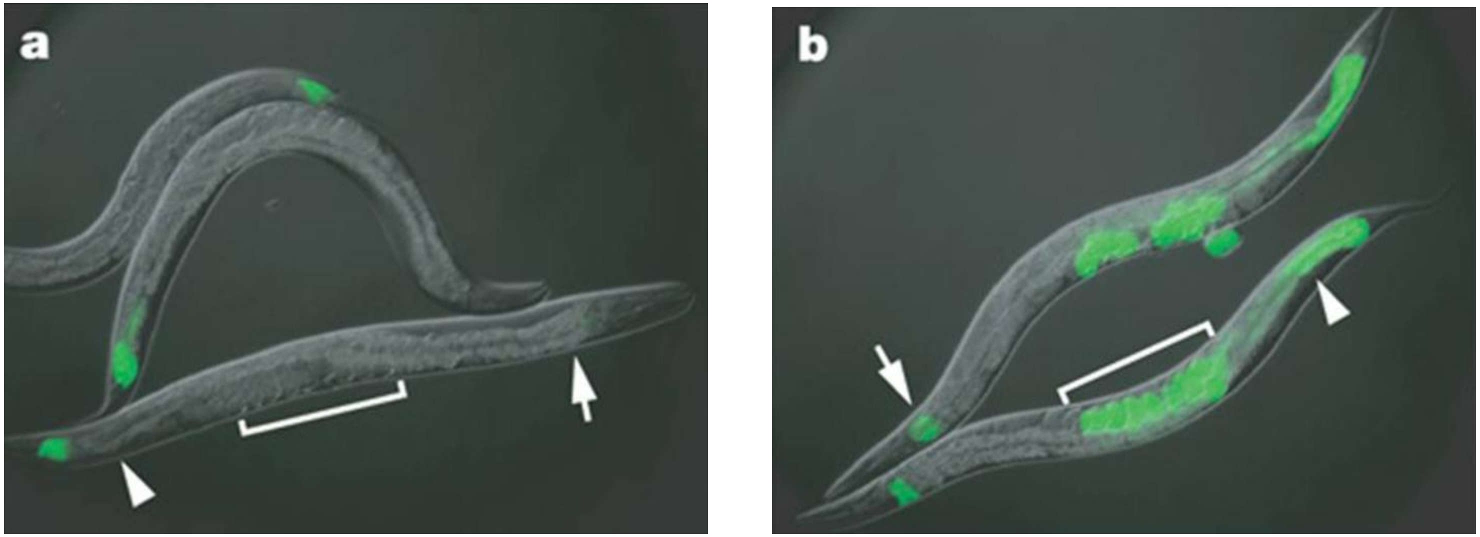
Could it be that dsRNA is the mediator in this silencing process ?

→

- dsRNA induces **potent specific interference** - a few dsRNA molecules are sufficient to silence the homologous gene's expression

1998 (Fire & Timmons)

- Silencing of a green fluorescent protein (GFP) reporter in C.elegans occurs when animals feed on bacteria expressing GFP dsRNA (a) But has no effect in animals that are RNAi deficient (b)



* Note that silencing occurs throughout the body of the animal, with the exception of a few cells in the tail that express some residual GFP

* The lack of GFP-positive embryos in (a) demonstrates the systemic spread and inheritance of silencing

1998 Tabara & Mello

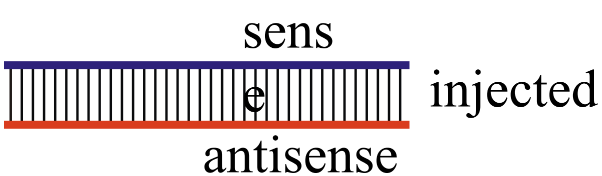
• Soaking the worms in dsRNA can also induce interference

→ Both soaking and feeding work in the same efficiency, but effects are less potent than those obtained by injection

Discovery of RNAi

Nature (1998) 391:806-811

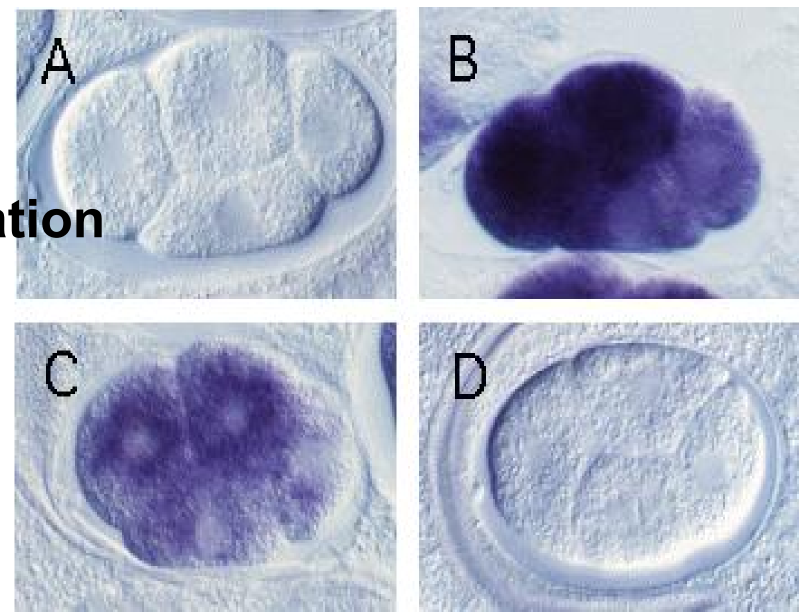
Double-stranded RNA



C. elegans

A, CK- ; B, Ck+, uninjected;
C, Antisense RNA; D, dsRNA

***In situ*
hybridization**



Conclusion: dsRNA reduced mex-3 mRNA better than antisense mRNA

The biochemical mechanism of RNAi -

1999 (Baulcombe & Hamilton)

- unable to detect PTGS inducing using anti-sense RNA molecules in similar size to typical mRNA

could shorter RNA molecules mediate the interference ?

↓

RNAs of ~25 nt are present in plants undergoing silencing but absent from plants not undergoing silencing

2000 (Zamore & Tushcl)

- role of siRNAs
- mRNA is <u>cleaved at 21-23 nt intervals</u>

↓

dsRNA provides not only specificity, but also determines mRNA <u>cleavage boundaries</u>

Why dsRNA ?

• The presence of dsRNA or other aberrant nucleic acid is “identified” as a viral infection

↓

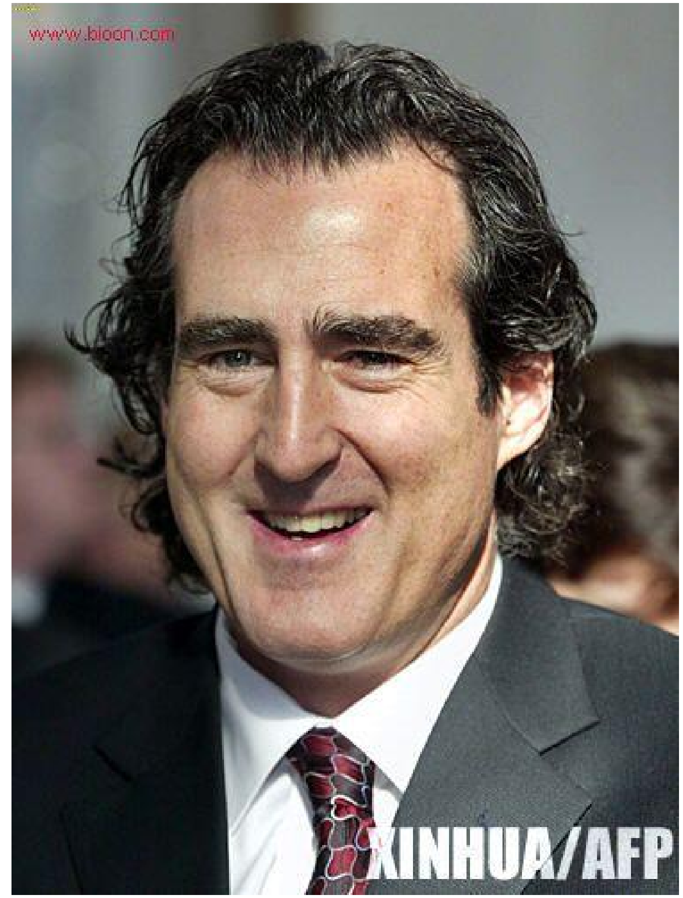
• “invasion” of RNA → put out of action any mRNA matching it

• No discriminations – if “invading” RNA mimics part of host’s genetic sequence – both host and viral genes are silenced

The Nobel Prize in Physiology or Medicine 2006

Andrew Fire

Stanford University,
California, USA

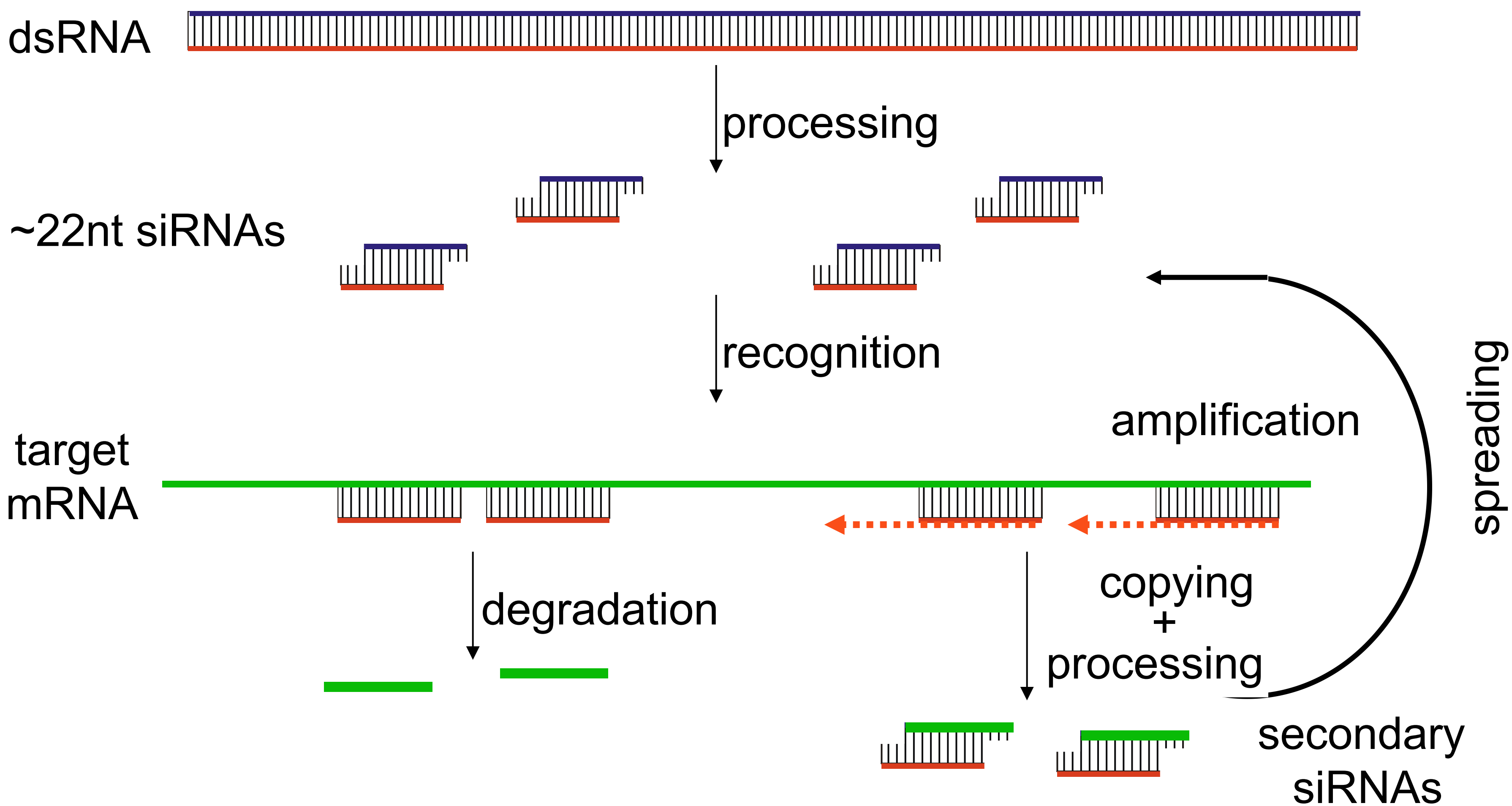


Craig Mello

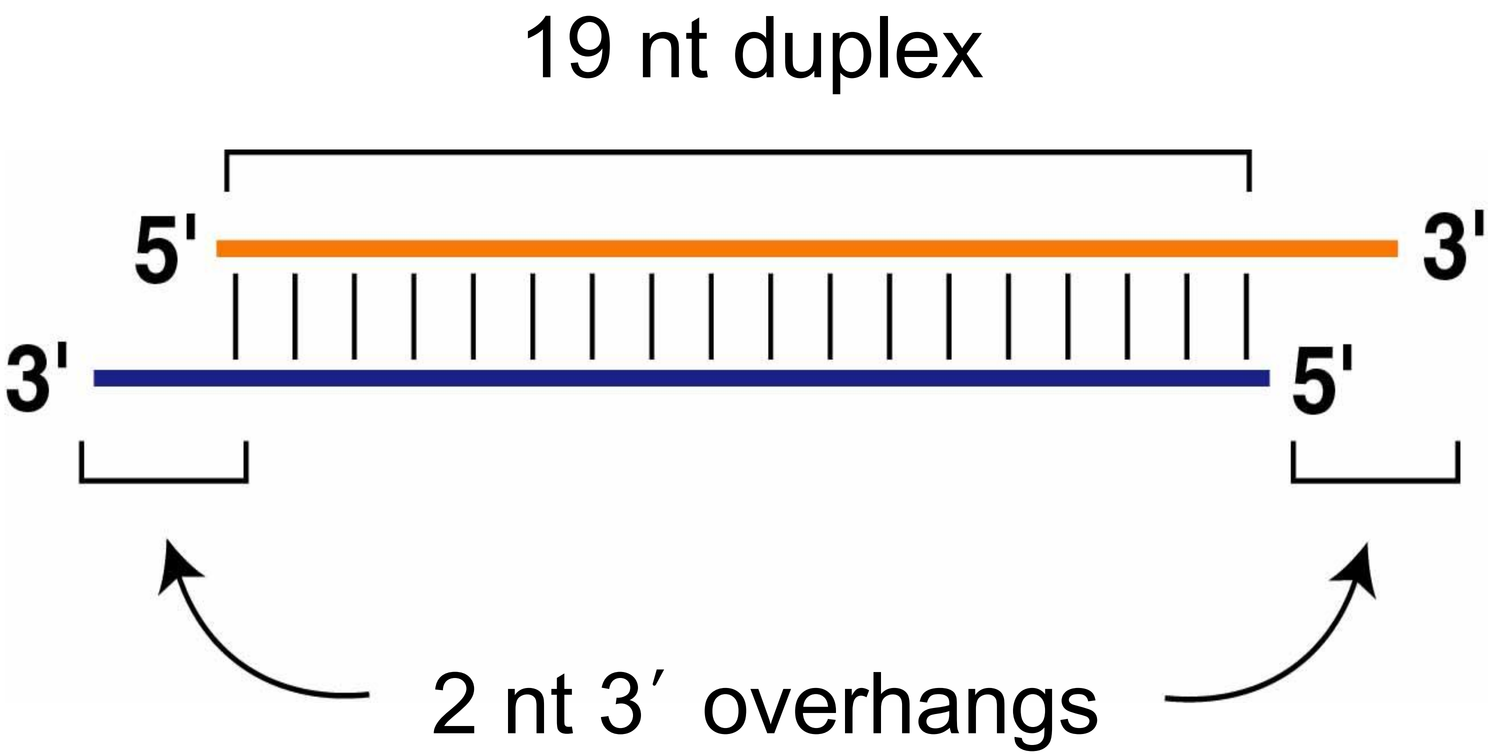
University of Massachusetts
Medical School in Worcester,
USA

Mechanism of RNAi

Gene Silencing directed by ~22nt RNAs



siRNAs have a defined structure

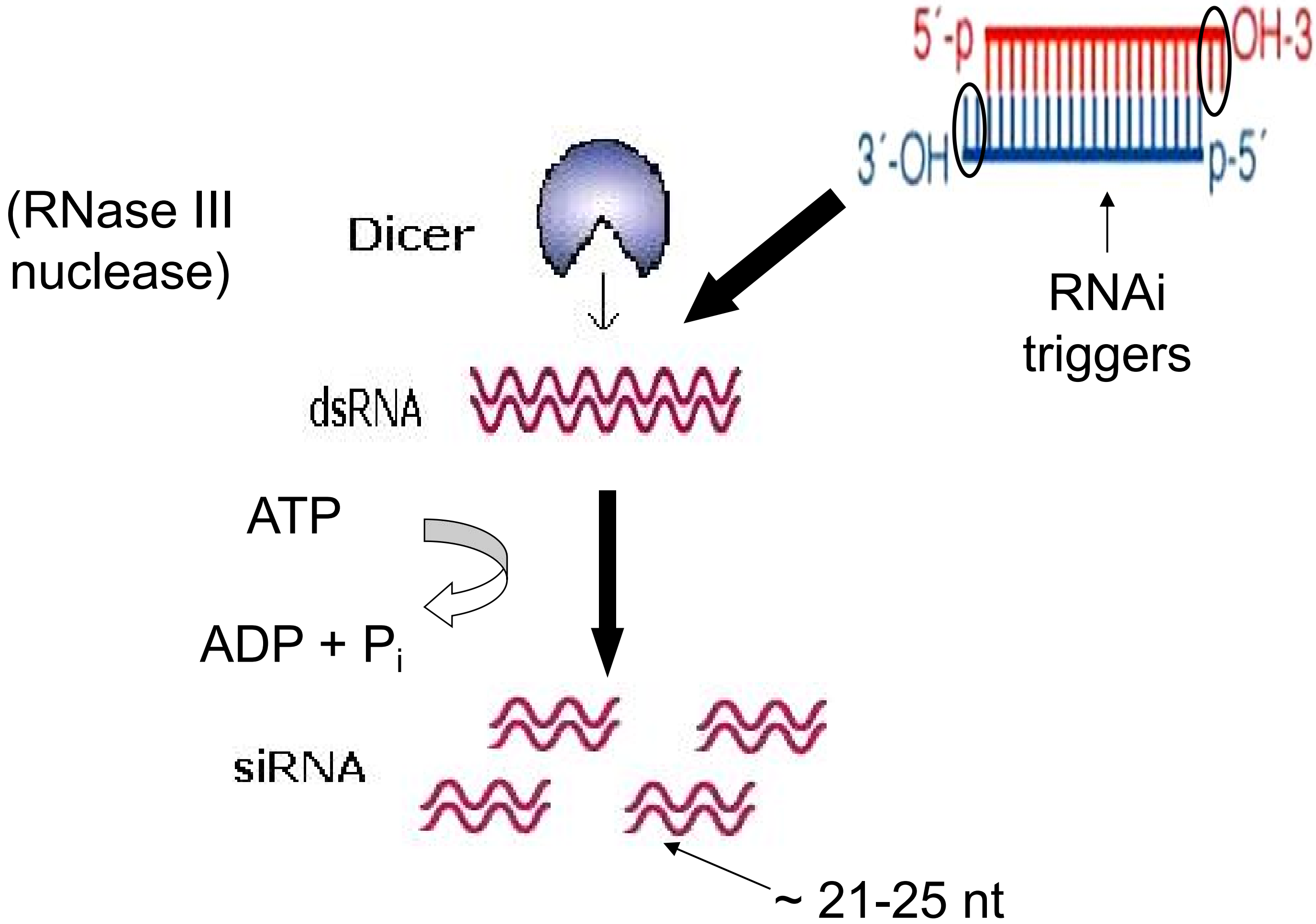


The Mechanism of RNA Interference

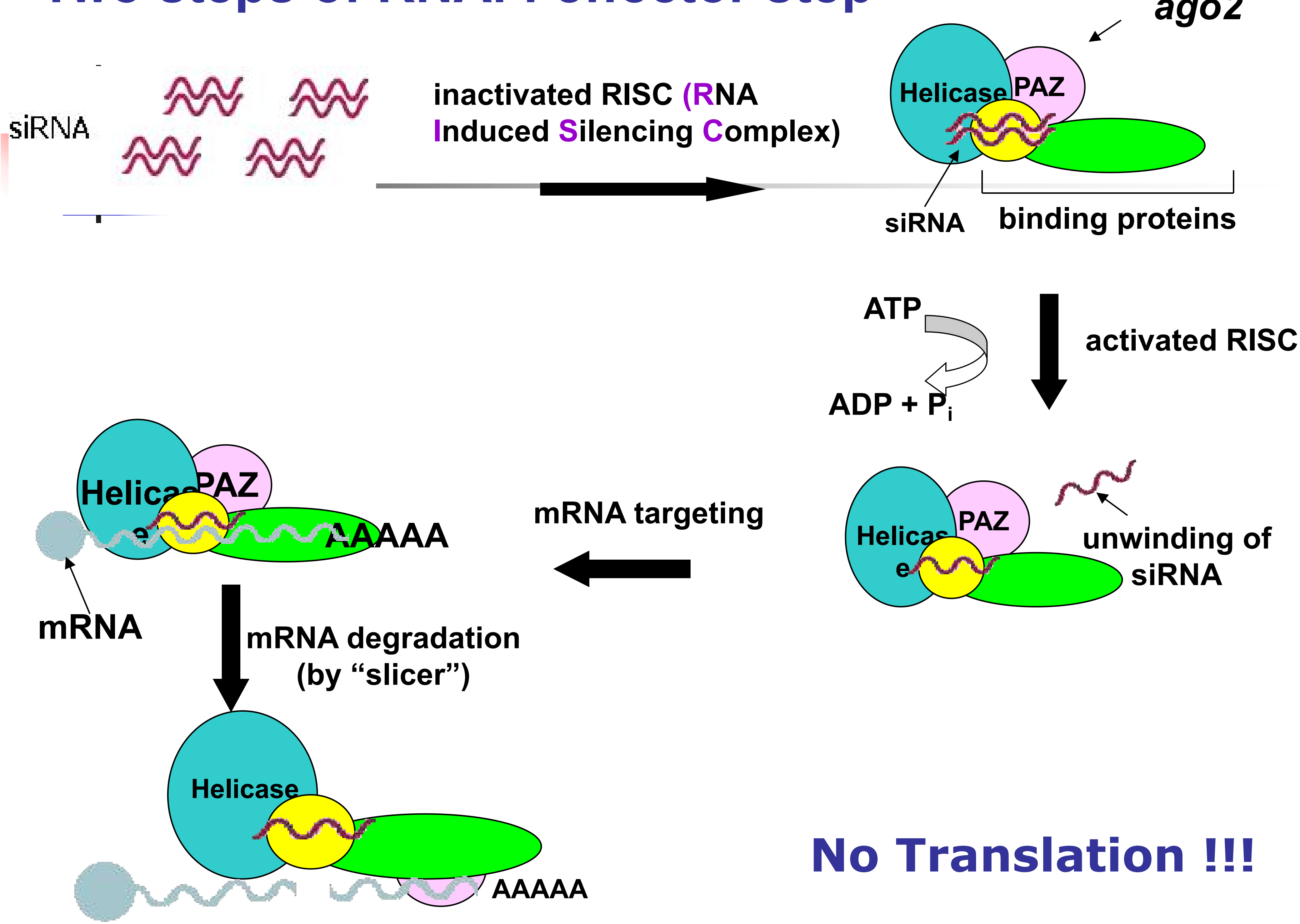
- **Initiation step**

- **Effector step**

Two steps of RNAi : Initiation step



Two steps of RNAi : effector step



No Translation !!!

How is it possible: a few molecules are sufficient for silencing vast amounts of mRNA ?

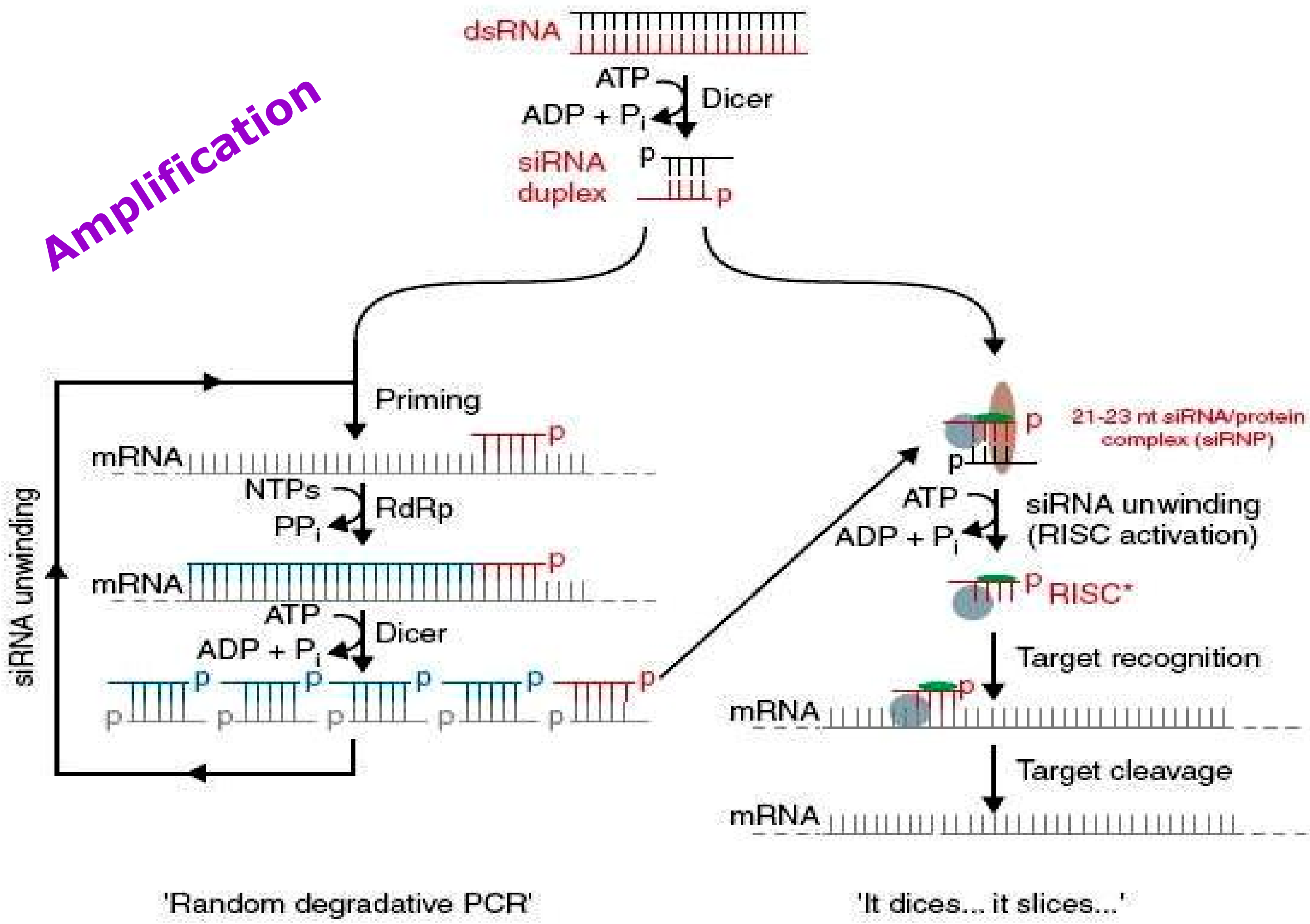
- **Each siRNA fragment can target the homologous mRNA**

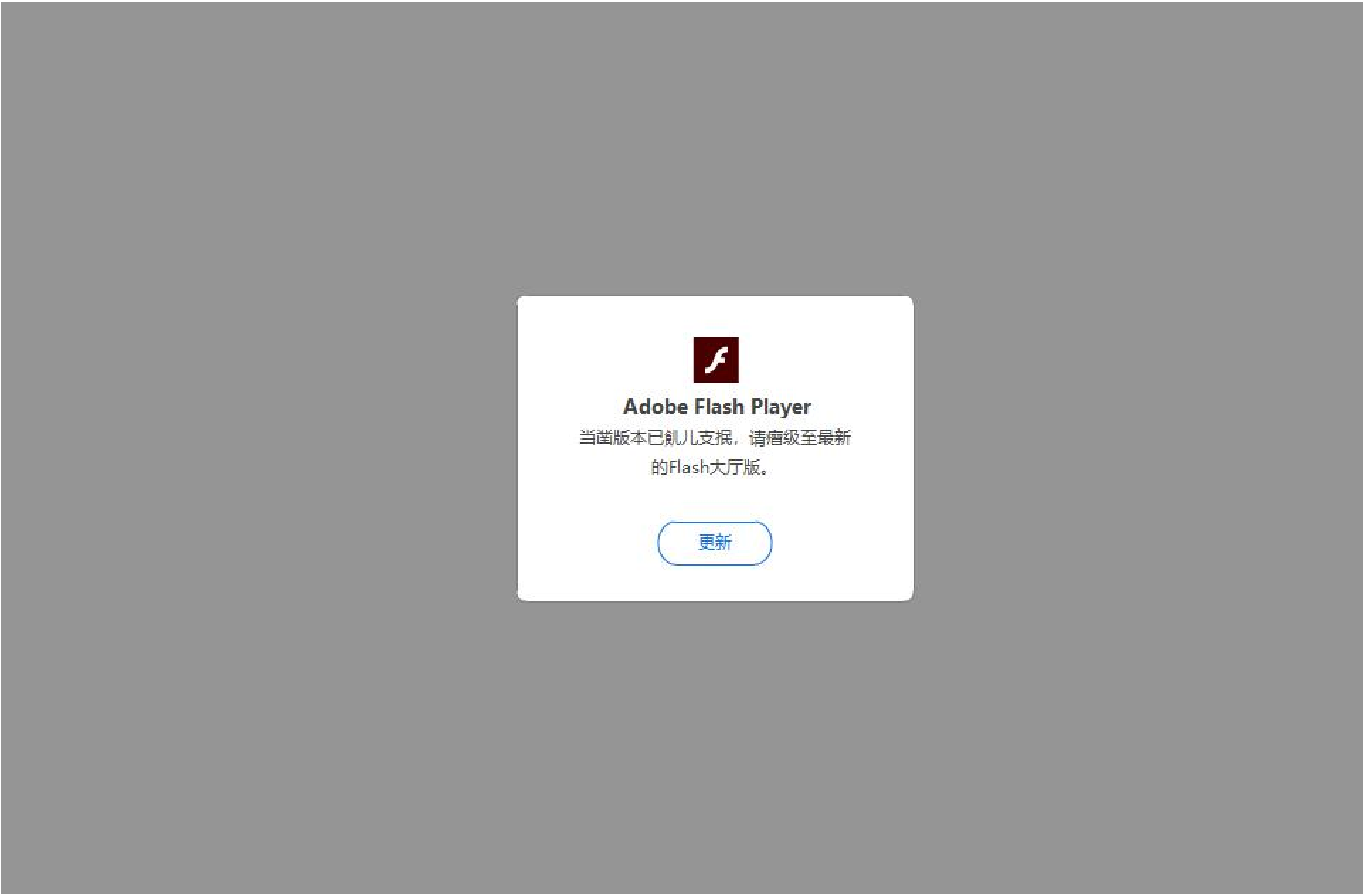
- **Catalytic mechanism: each siRNA fragment can be used several times**

Random degradative PCR [Lipardi et al, 2001]

- via RNA dependent RNA Polymerase (RdRp)

- siRNA primers convert mRNA into dsRNAs that are degraded to generate new siRNAs.

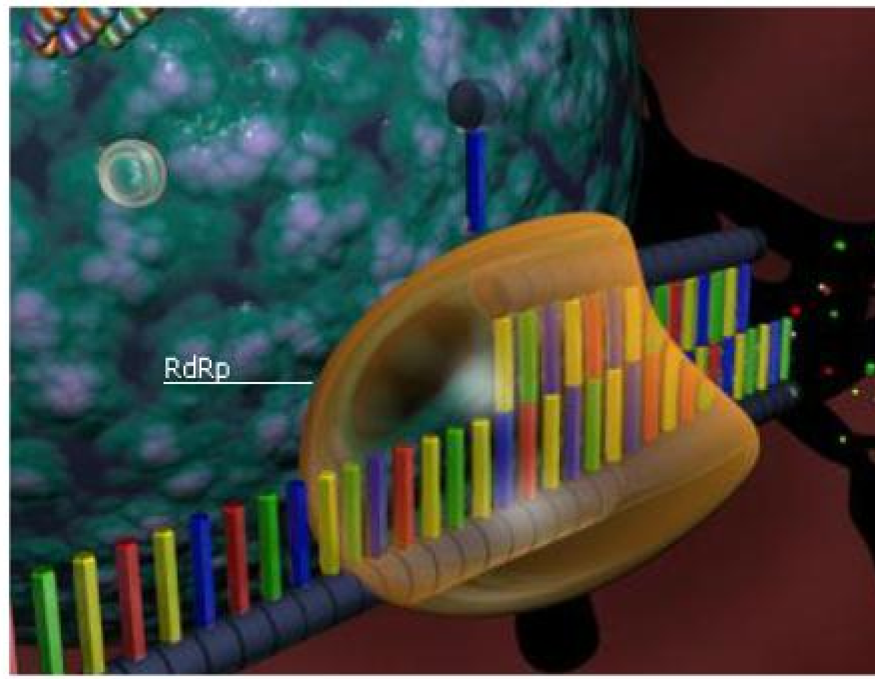




Adobe Flash Player
当菌版本已飢儿支抿，请瘤级至最新
的Flash大厅版。
更新

In plants, the aberrant RNA that results from RISC-mediated cleavage can also serve as a template for RNA-dependent RNA polymerase (RdRp) to make a new dsRNA molecule. This process relies on unprimed RNA synthesis in which the aberrant RNA is used as a template.

Part 3: RNAi amplification



Role of the RNAi machinery in heterochromatin formation and gene silencing

Target a gene’s control region through DNA and histone methylation

Biological Significance of RNAi

- Most widely held view is that RNAi evolved to protect the genome from viruses (or other invading DNAs or RNAs)

- In the past decades, very small (micro) RNAs have been discovered in eukaryotes that regulate developmentally other large RNAs

Endogenous RNAi inducing genes -

1993 (Lee et al)

- gene *lin-4* (non coding ~22 nt) encodes small RNAs with antisense complementarity to *lin-14 3' UTR*
- negatively regulates *lin-14* translation

For 7 years no other small RNA was discovered

2000 (Reinhart et al)

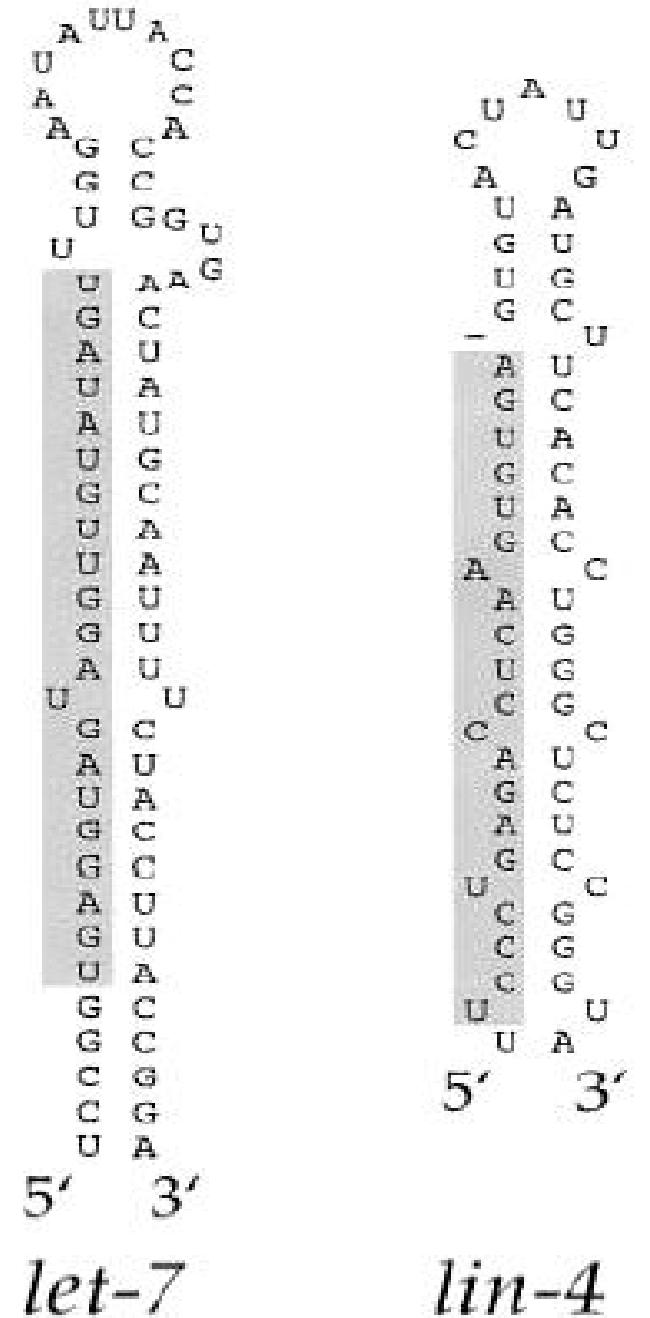
- gene *let-7* RNA (21 nt) regulates developmental timing in *C. elegans*
- negatively regulates *lin-14* translation (same as lin-4)

⬇

A new **small RNAs** specie :miRNA

MicroRNA (miRNA) :

- Non coding small RNAs
- Regulate expression via complementary base-pairing
- unlike siRNAs, miRNAs may down-regulate expression after translation initiation without affecting mRNA stability
- stem-loop structure is highly conserved



miRNA biogenesis

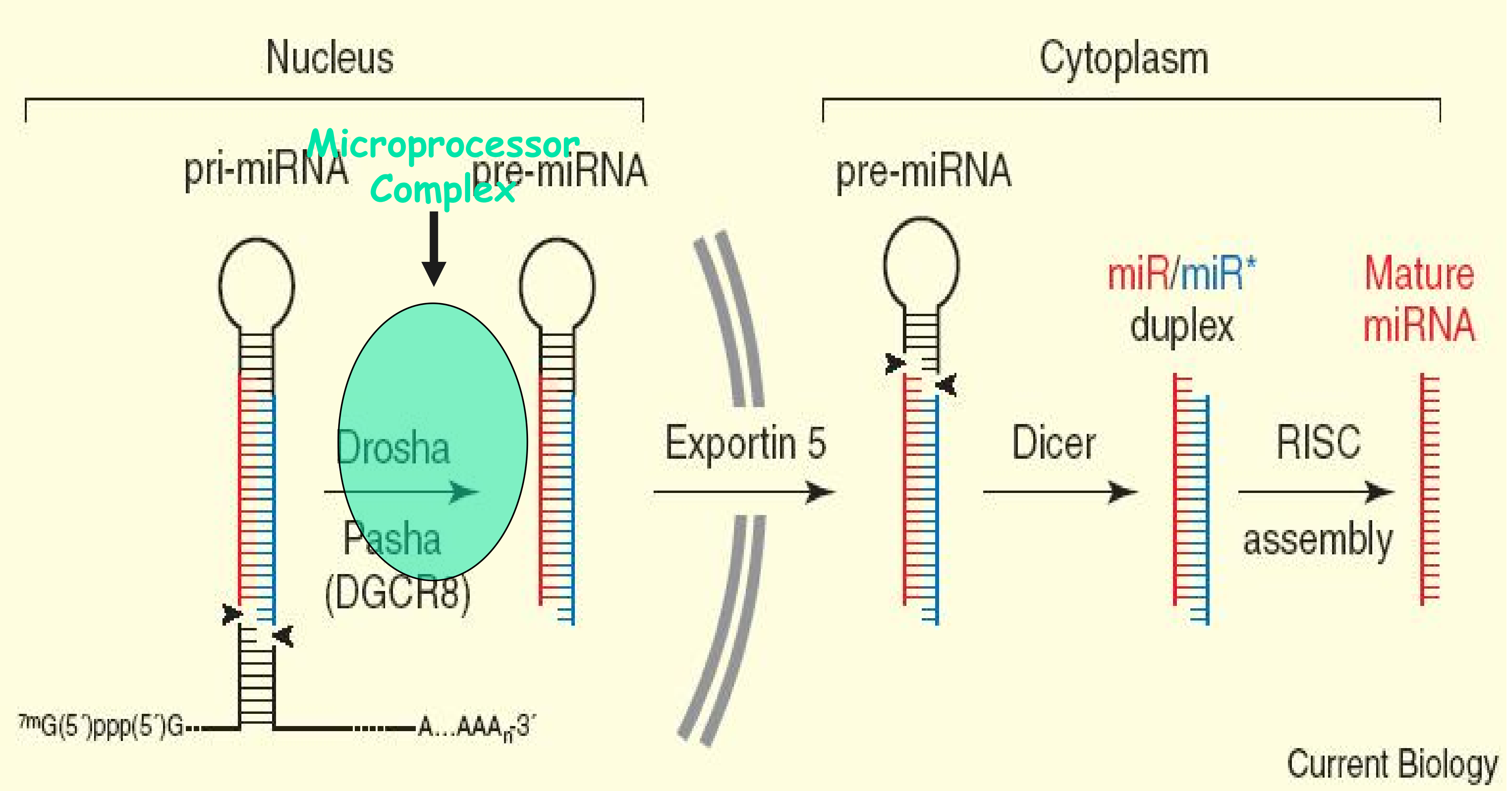
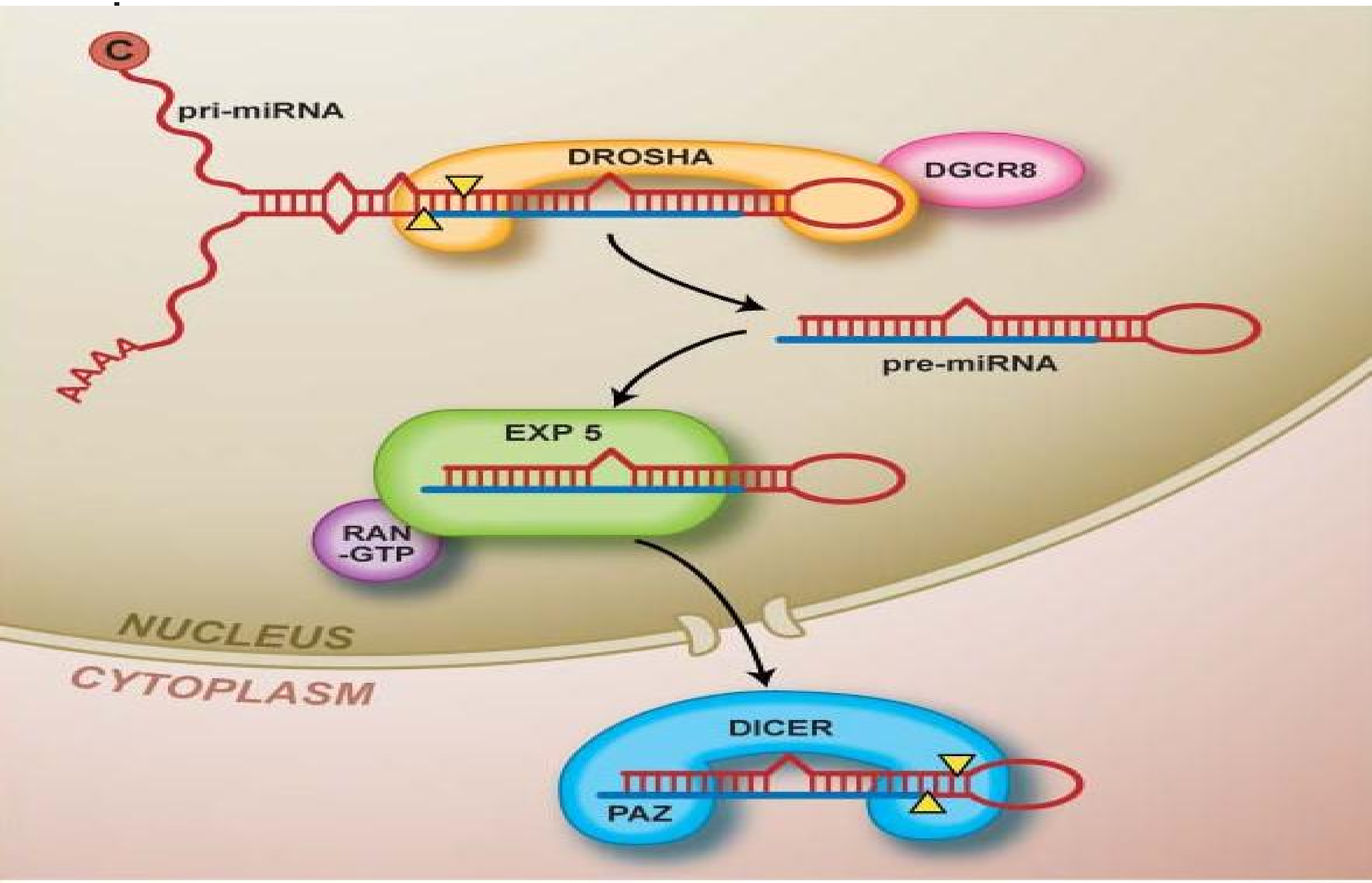
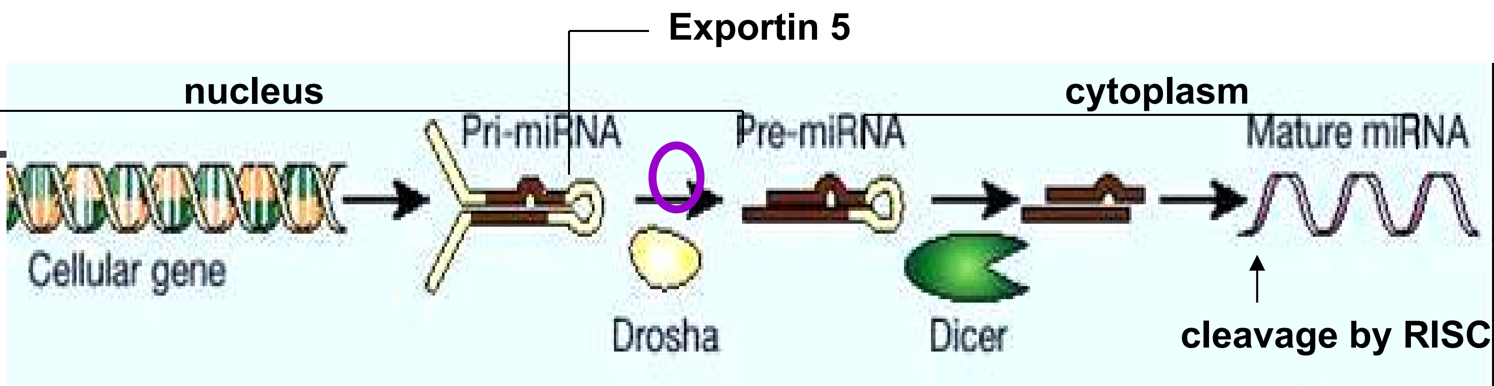


Illustration of miRNA processing



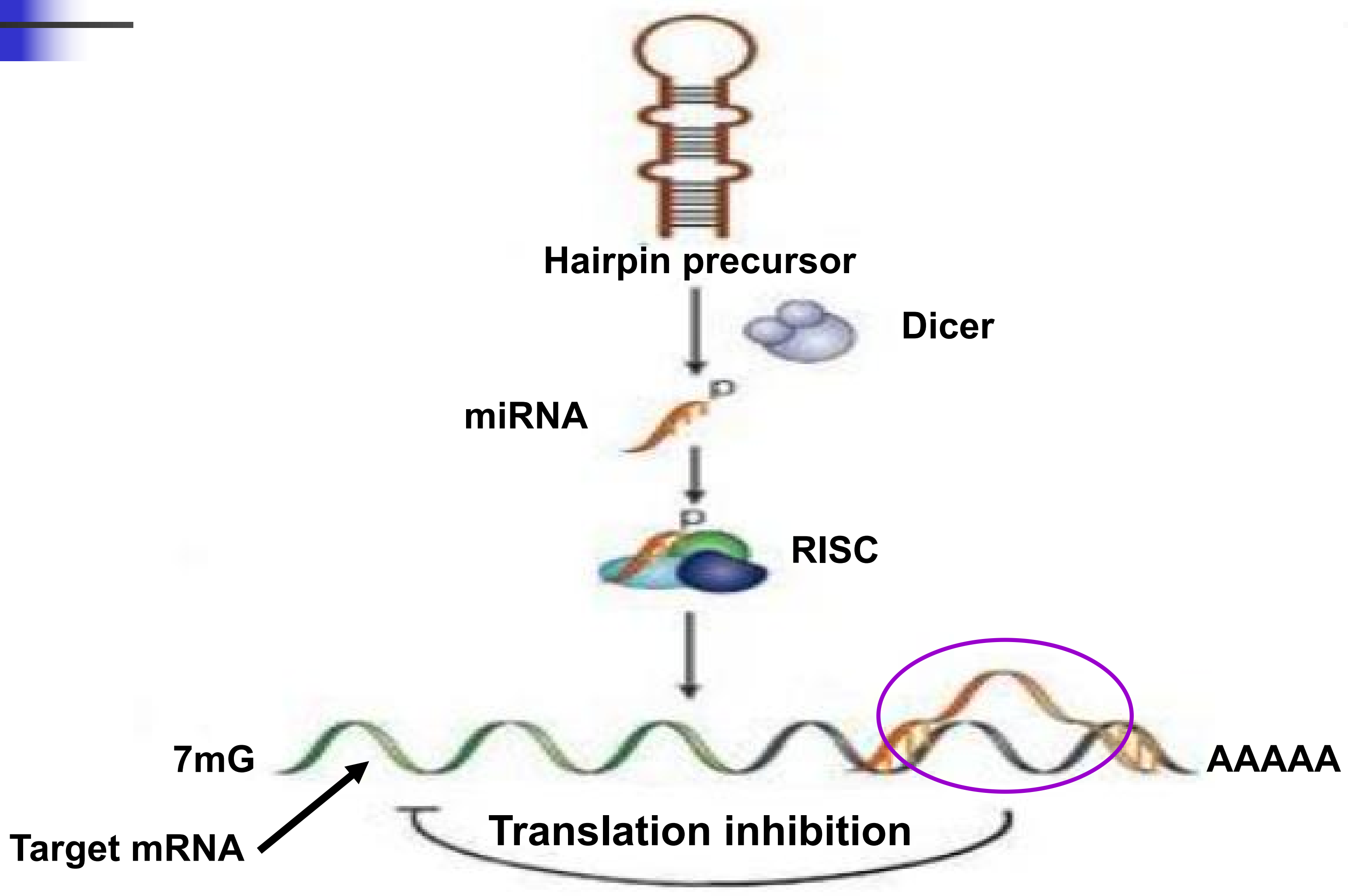
miRNA biogenesis



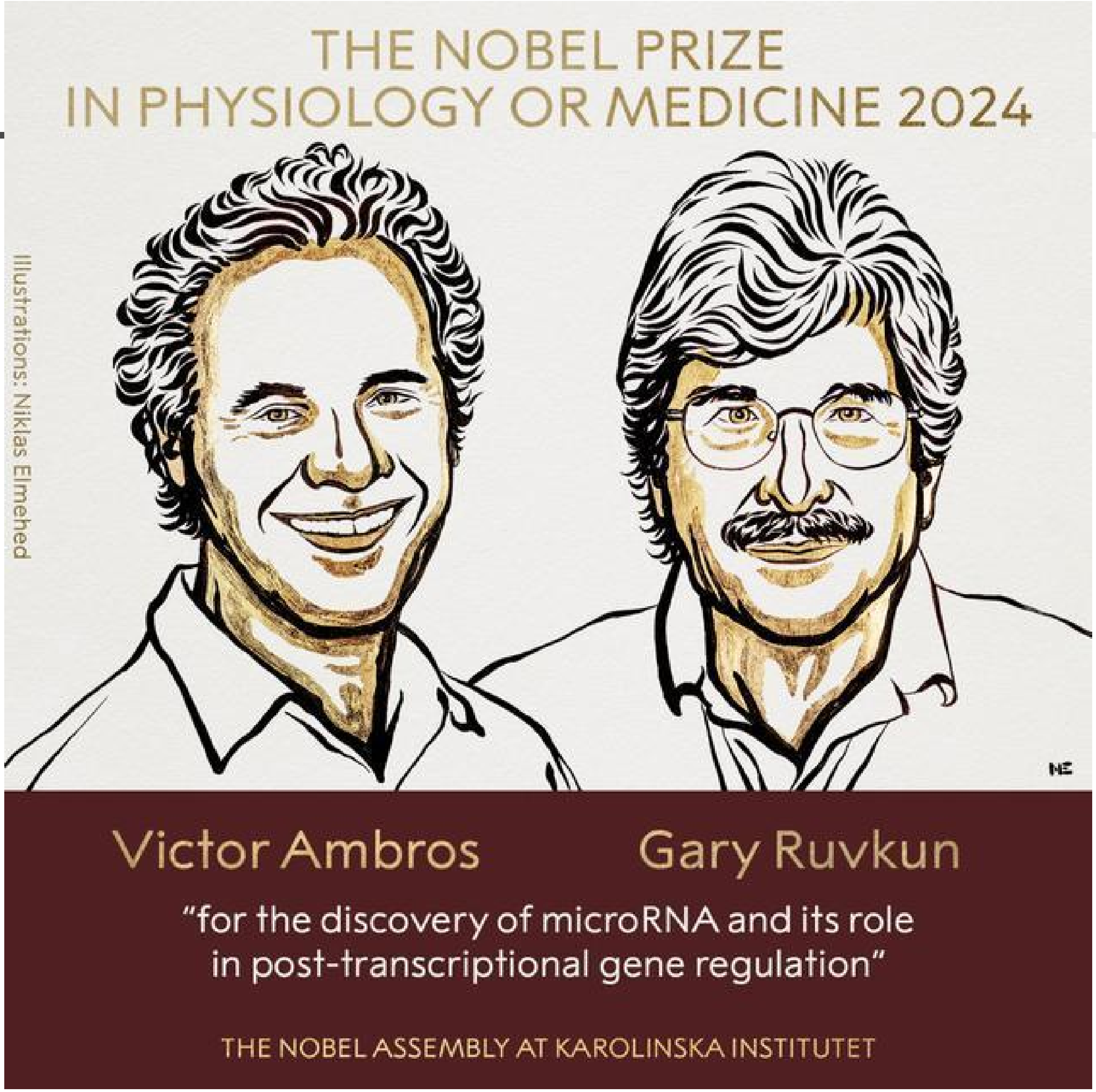
- Transcribed from endogenous gene as **pri-miRNA**
  - Primary miRNA: long with multiple hairpins
  - Imperfect internal sequence complementarity
  - pri-miRNA is identified by drosha due to its high affinity to hairpin loops
- Processed by the RNase III enzyme Drosha into a ~70 nt hairpins → Pre-miRNA
- Pre-miRNA is exported to the cytoplasm by **Exportin 5**

  cleaved by **Dicer** into the mature miRNA
  - Symmetric 2nt 3’ overhangs, 5’ phosphate groups

miRNA pathway



2024年诺贝尔生理学或医学奖

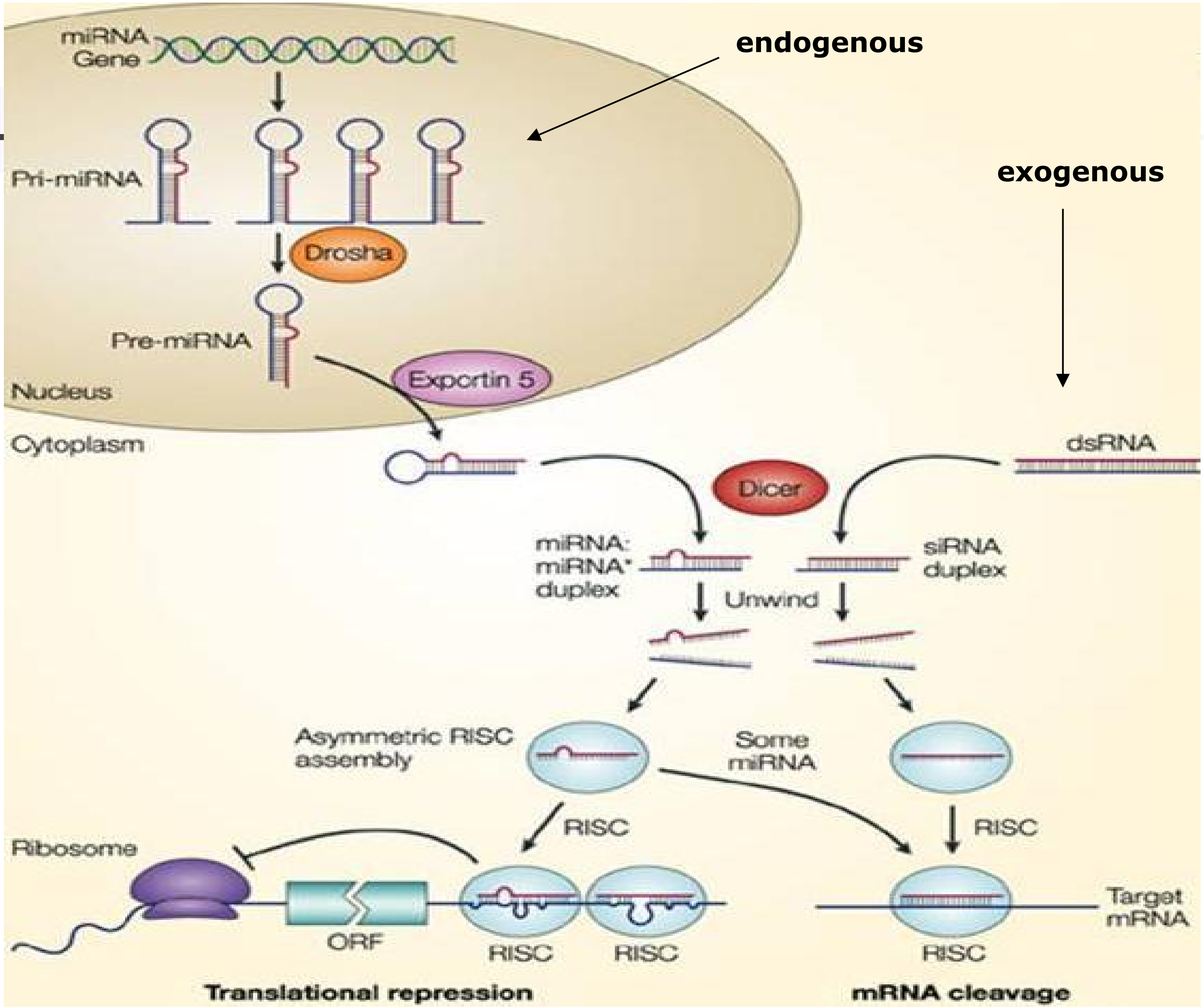


What are the functions of miRNA?

- Involved in the post-transcriptional regulation of gene expression
- Important in development
- Metabolic regulation
- Multiple genomic loci

siRNA vs. miRNA

- mRNA cleavage vs. Translational Repression



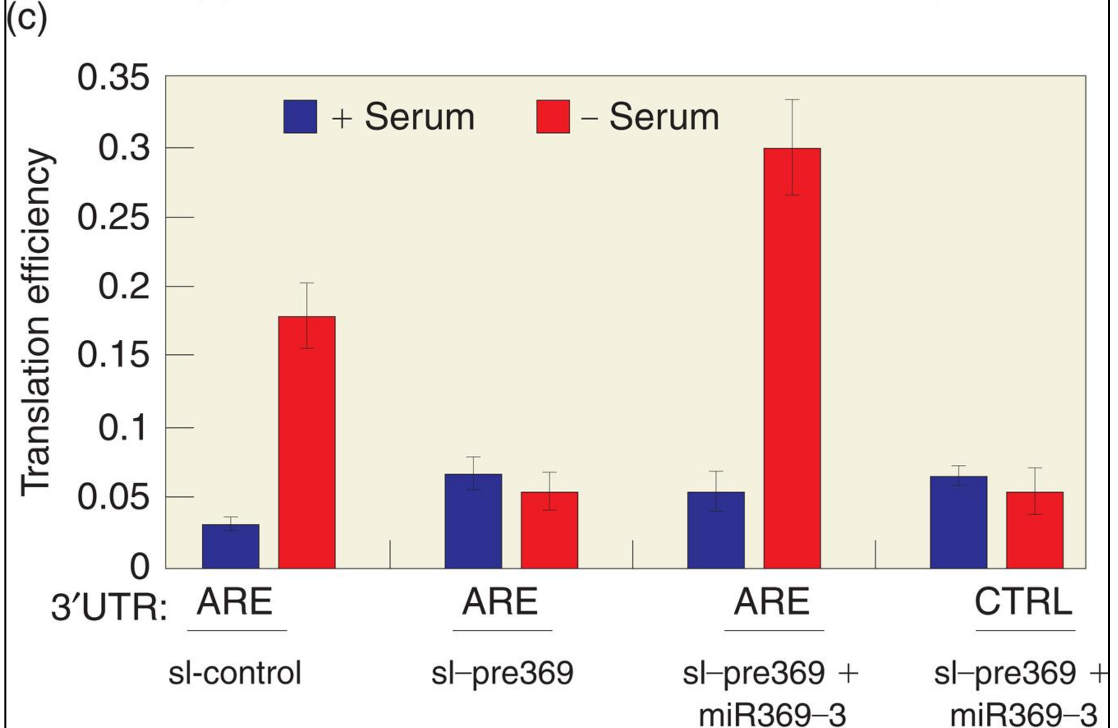
Mechanism of miRNA proposed

- microRNAs were acting in some way to inhibit gene translation into protein
- Some microRNAs closely match their sequences against particular messenger RNA sequences also target them for destruction
- Detailed mechanism is still in discovering.

MicroRNA can also activate translation

miR369-3, with the help of AGO2 and FXR1, activates translation of TNFα mRNA in serum-starved cells.

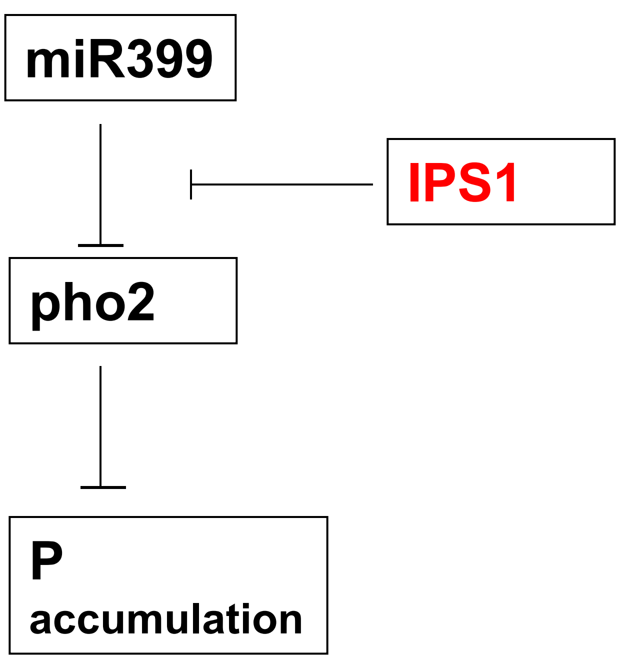
Fig. 16.41c



ARE: AU-rich element

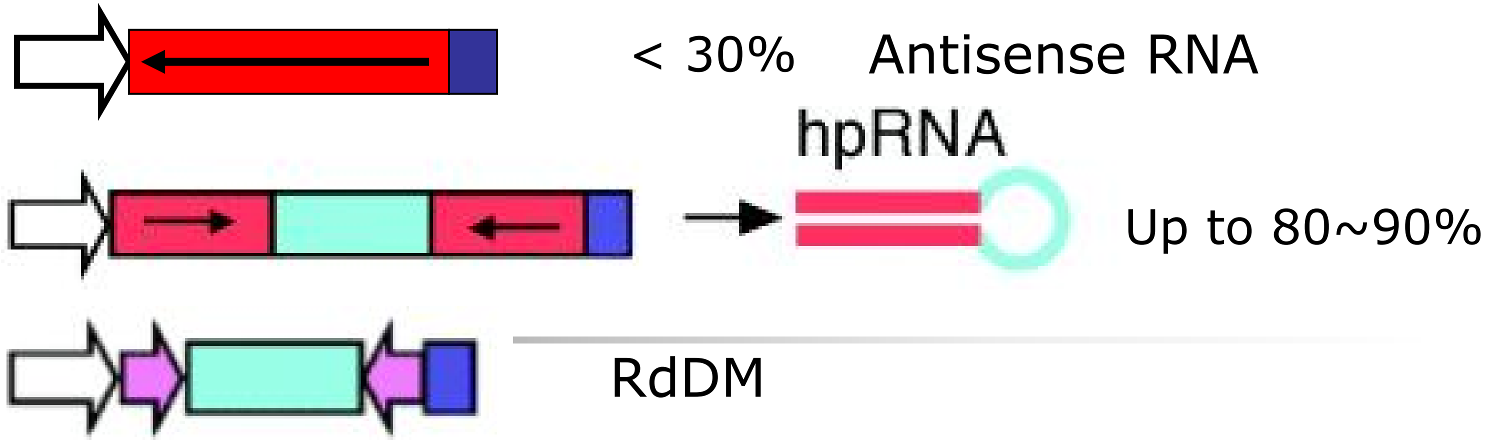
Long non-coding RNA (lncRNA)

- Non-protein coding transcripts longer than 200 nucleotides
- Some function in regulation of gene transcription
- Some function in translation

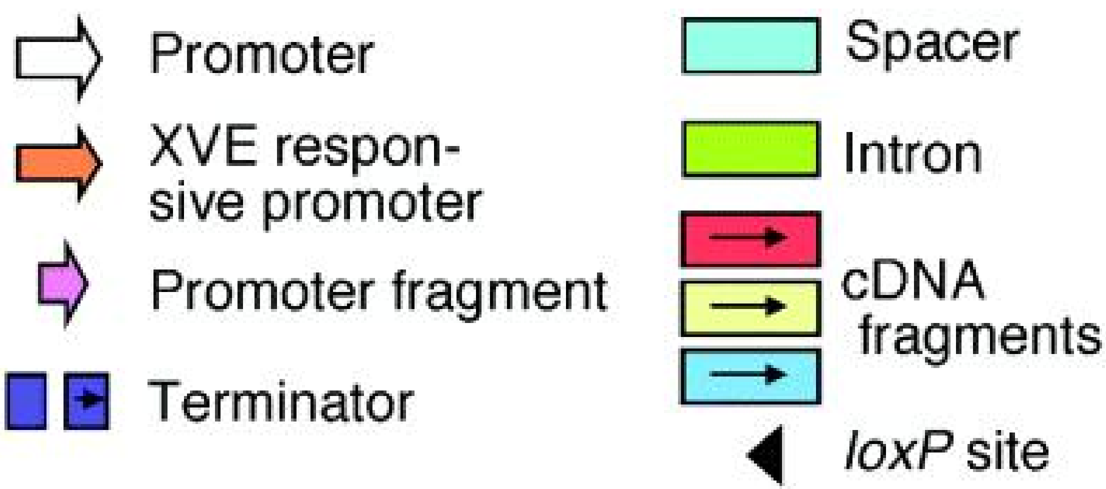


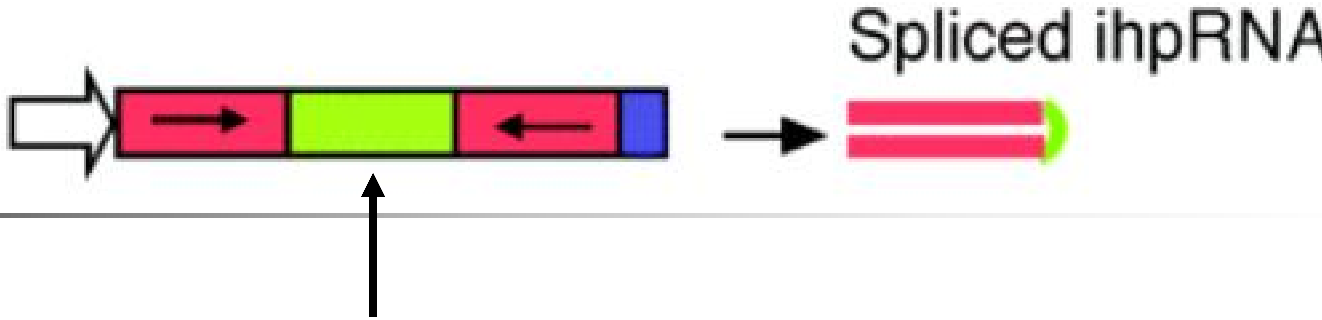
RNAi Applications

- **To protect the genome from viruses (or other invading DNAs or RNAs)**
  - Antiviral Defense
  - Suppress Transposon Activity
  - Response to Aberrant RNAs
- **Genetic tool:** Probing Gene Function
- **Gene therapy:** Combat Viral Infection; Treat Genetic Diseases (New expression strategies)

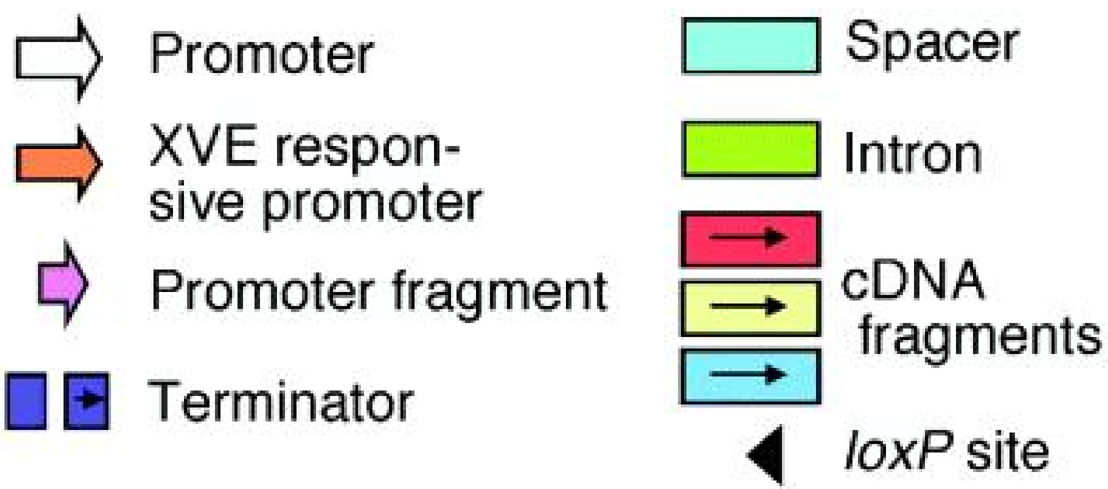


A promoter fragment is used instead of a cDNA fragment, the resultant hpRNA directs DNA methylation of the homologous promoter sequence, which results in transcriptional gene silencing (Mette et al., 2000).

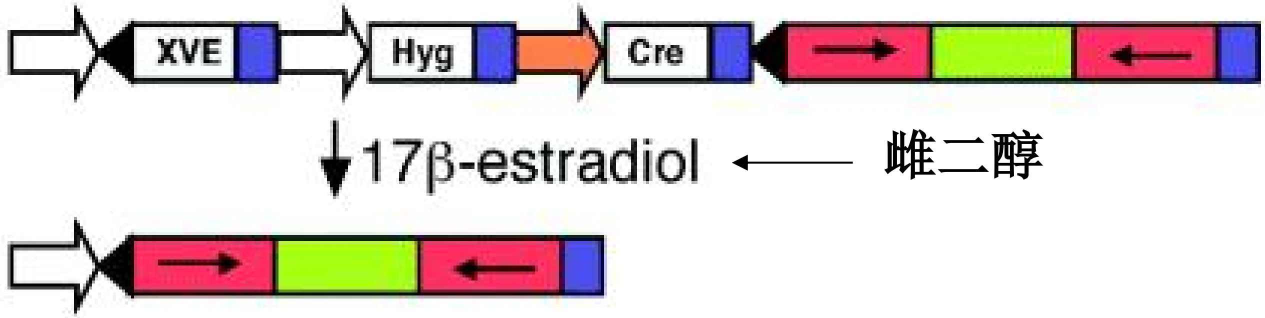




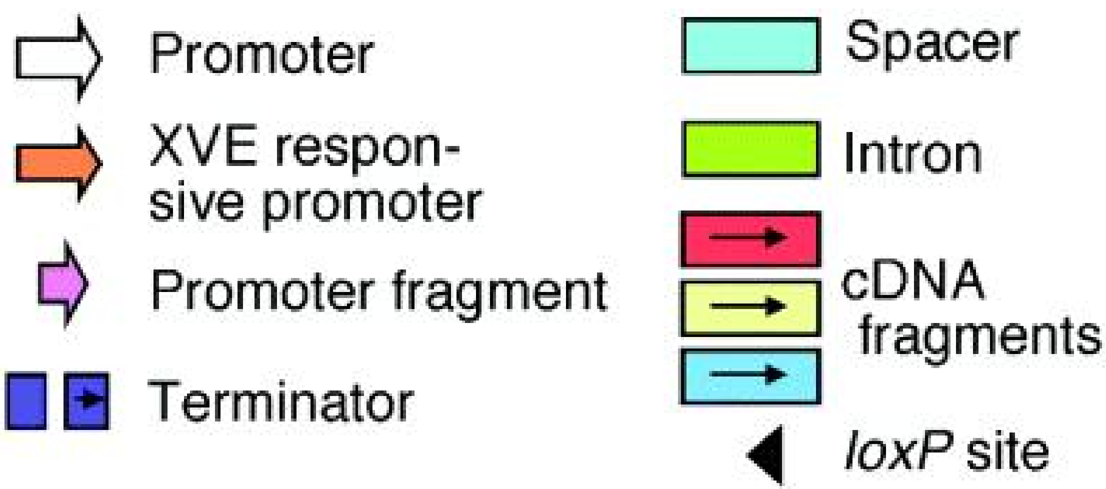
The use of an intron as a spacer (intron hairpin RNA, ihpRNA) further increases the silencing efficiency to nearly 100% of transformants.



Inducible and Controllable RNAi



This system is suitable for functional analysis of essential genes or genes that have multiple roles at different stages of plant development.

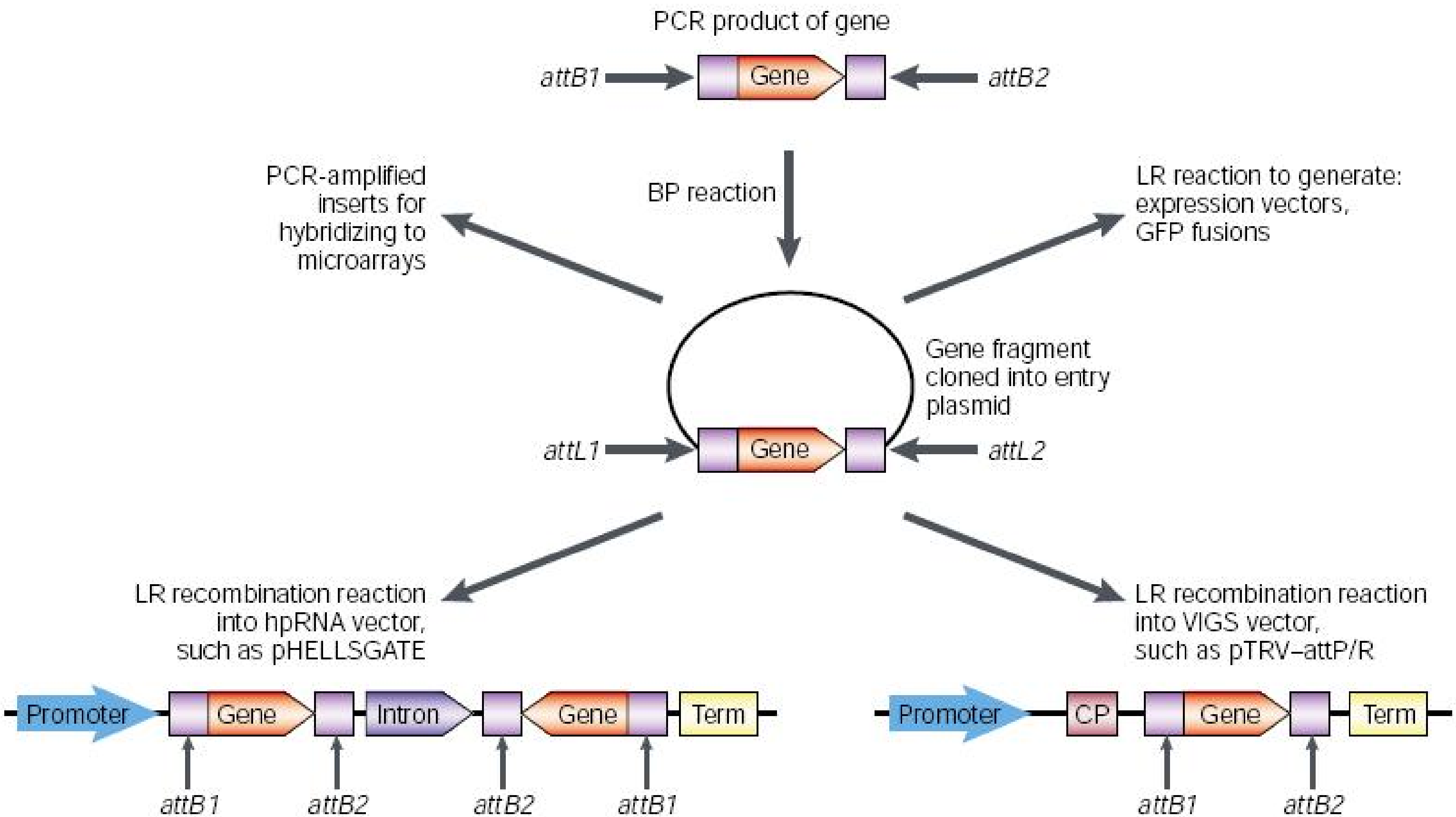


- However, the cloning of two DNA fragments into a single vector in the correct orientation is labor intensive and inappropriate for high-throughput analysis.

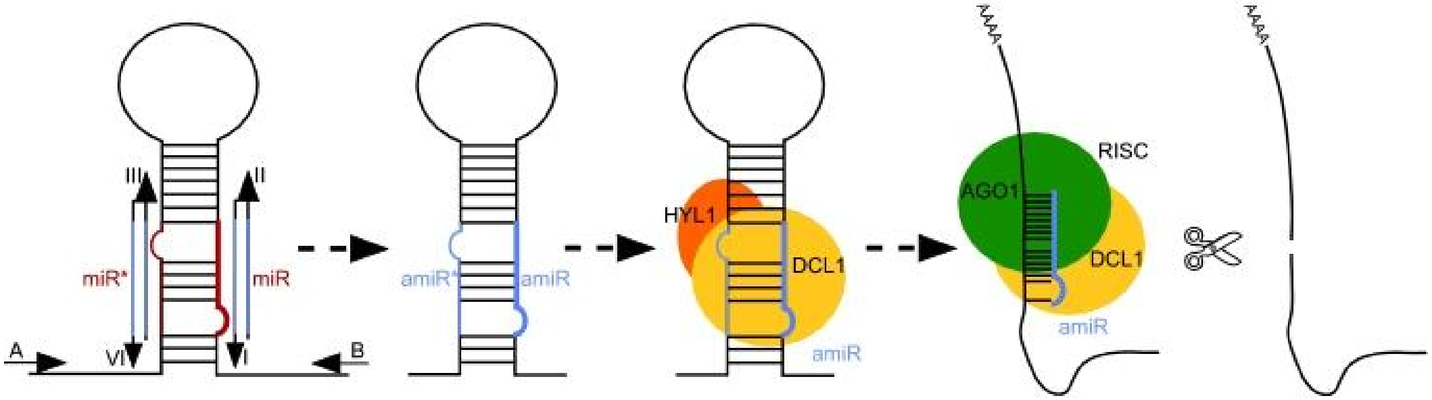
- In addition, the large binary vector size (about 10 kb) limits the restriction sites available for cloning a DNA fragment and results in low transformation efficiency.

How to……?

Gateway System Used in RNAi

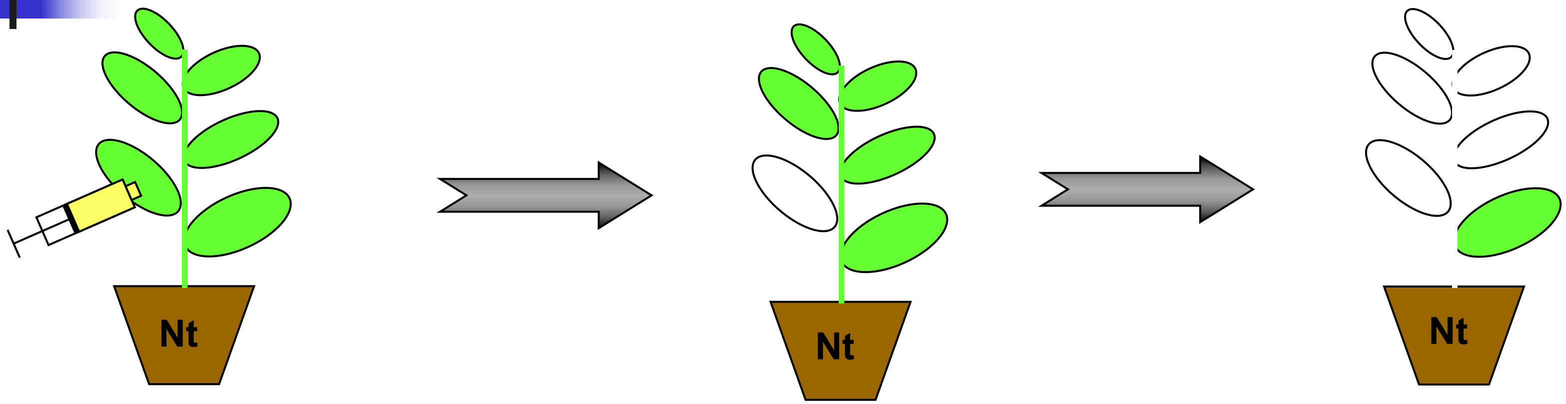


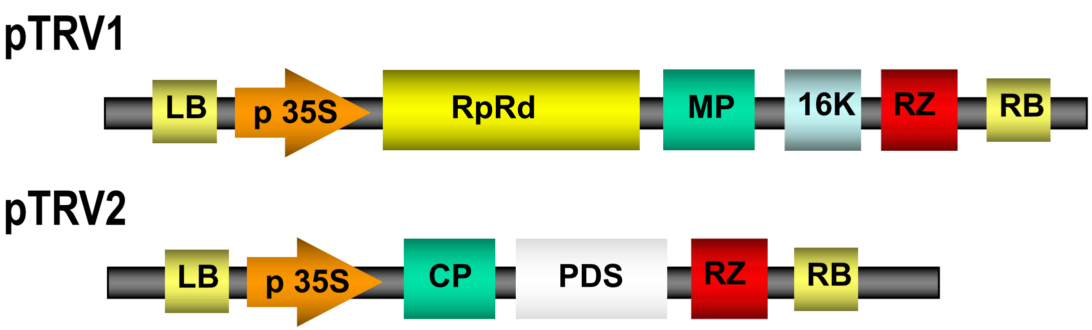
Artificial miRNA



http://wmd2.weigelworld.org/cgi-bin/mirnatools.pl

Virus-induced gene silencing (VIGS) **system**





Using RNAi technology, gossypols (棉酚)-free cotton has been developed

Application of RNAi technology

- **Advantages**
  - gene specific
  - High gene silencing efficiency
  - Screening targeted plants takes less time
  - Highly inducible (heat shock gene promoter)
  - Fast
- **Disadvantages**
  - It does not knockout a gene for 100%
  - siRNA tends to activate unwanted pathways

    --Expensive

    --Ethical problems

Improvement of RNAi based methods

High-throughput

Convenience

Cheap

Genomic

inducible

mRNA Degradation and Translation Repression, in P-Bodies

- Processing bodies (P-bodies or PBs) are discrete collections of RNAs and proteins that are involved in mRNA decay and translational repression.
- P-bodies are cellular foci where mRNAs are destroyed or translationally repressed
- These cellular foci are enriched in enzymes that deadenylate mRNAs, decap mRNAs and catalyze 5’ to 3’ degradation of mRNAs
- Degradation of mRNA occurs by a non-RNAi-like mechanism

Relief of Repression in P-Bodies

- Although many mRNAs are degraded in P-bodies, many others are held and repressed there
- In a liver cell line (Huh7), translation of the CAT-1 mRNA is repressed by the miRNA miR-122 and the mRNA is sequestered in P-bodies
- Upon starvation, the translation repression of the CAT-1 mRNA is relieved and the miRNA migrates from P-bodies to polysomes
- This derepression and translocation of the mRNA depends on the mRNA-binding protein HuR and on its binding site in the 3’-UTR of the mRNA

Student's presentation (today)

- Paper: Maninjay et al. A Long Noncoding RNA lincRNA-EPS Acts as a Transcriptional Brake to Restrain Inflammation. 2016, Cell 165, 1672–1685
- Presenter: Group 10